# bb-adc — Design Document

## PCB-0014-A

hwde v1 pipeline

generated 2026-09-25 18:09:29

## Contents


**1 Board and Run Metadata** **1**

**2 Overview** **1**

**3 Requirements** **1**

**4 Architecture** **8**

**5 Schematic** **40**

**6 Layout** **42**

**7 Verification** **42**

**8 DFM and Fabrication** **48**

**9 Run Record (Appendix)** **49**

**10 Artifact Index** **56**


# 1 Board and Run Metadata

| Field | Value |
|---|---|
| board | bb-adc |
| part number | PCB-0014-A |
| workspace | `boards/PCB-0014-A_bb-adc` |
| phase | P9 |
| gate status | all 6 recorded gates pass |
| state created | 2026-08-16T17:25:12 |
| state updated | 2026-08-20T00:46:26 |

# 2 Overview

learning block-basics: a single-channel analog-to-digital converter. Analog input 0-5 V DC on a screw terminal; 12-bit or better; up to 10 kSa/s; DC accuracy matters more than speed. Digital output to a host over a 0.1 inch header, which also feeds the board its 3.3 V rail. Workspace boards/bb-adc.

—

Owner brief, verbatim, received 2026-08-16. The text above is the owner's own wording and is the authoritative statement of the task; nothing below it is part of the brief.

Mode token resolved at P0 by `state.py mode`:

- token: `learning block-basics:`
- target: `block-basics` -> scope tier `block-only`, binding `canonical`

- geometry: OUTPUT (the placement earns the outline; `board_init --outline auto`)
- stage under study: none

### Stackup

Chosen stackup: `JLC2313_1.6`

## 3 Requirements

### Requirements: bb-adc

#### 1. Function

A single-channel analog-to-digital converter, built as a study article. It takes one DC analog voltage in the range 0 to 5 V on a screw terminal, digitizes it to 12 bits or better at a rate up to 10 kSa/s, and hands the result to a host over a 0.1 inch header. That same header feeds the board its only supply, 3.3 V. DC accuracy is the dominant requirement; conversion speed is secondary and explicitly subordinate to it. Nothing else is on the board: no MCU, no display, no indicators, no second channel.

BUILD MODE: the brief opens with the token `learning block-basics:`. Resolved at P0 (`state.py mode`, recorded as a P0 decision; contract in `reference/build-modes.md`):

- mode / target: **block-basics**
- scope tier: **block-only**
- binding level: **canonical** - board geometry is an **OUTPUT** of the design, not an input to it
- stage under study: **none** (the block end to end, no single stage frozen as a bench fixture)

What that means here. The block under study is the ADC signal chain itself: the converter, exactly the support parts its datasheet requires for correct operation at the stated operating point, one input interface (the screw terminal) and one output interface (the host header), plus what the fab needs to build the board and the bench needs to hold it. Protection of every kind, filtering the datasheet does not require, indicators, config straps, test points beyond the block's own measurement need, any second rail or second IC the block does not need in order to work, and mechanical/enclosure features beyond mounting the bare board are OUT by MODE, not by engineering judgment - and must never be recorded as generally unnecessary. The mode's defaults are applied below (quantity 5, JLC PCBA single-sided, bench 0-50 C indoor, no enclosure, fewest honest layers, geometry earned by the placement, stop at P9) instead of being asked. The mode relaxes geometry, cost and packaging only: every electrical spec, every gate, the coverage checks and the research they trigger, DFM, the datasheet's own requirements and every safety question in section 8 stand unchanged.

**Recorded requirement tension, NOT solved here (P1 owns it).** The stated analog input span (0 to 5 V) sits ABOVE the board's only stated rail (3.3 V from the host header). Every way out of that is an architecture decision - which converter, which reference, whether the input is attenuated ahead of the converter, whether the block honestly needs something the 3.3 V rail cannot give it - and the architect decides it, not this document. Two consequences that must travel with the tension:

- It touches a MODE BOUNDARY. `block-only` excludes "any second rail the block does not need in order to work"; if P1 concludes the block DOES need one to work at the stated operating point, that is support, not a feature, and the call belongs at H1 with the owner, recorded as a decision.
- It interacts with the accuracy answer (Open question 2). Whatever sits between the terminal and the converter is inside the DC error budget, so the topology choice and the accuracy number have to be settled together.

## 2. Interfaces

Exactly two external interfaces, which is what the scope tier allows and what the brief states.

- **J1 - analog input.** Screw terminal (STATED). ASSUMED: 2-pole through-hole terminal block, 5.08 mm pitch - signal and its ground return - rated far above anything this board carries; pitch was not stated and 5.08 mm is the common, well-stocked JLC/LCSC choice that accepts bench lead wire. ASSUMED: single-ended input referenced to board ground, since the brief describes one channel of 0-5 V DC with no mention of a floating or differential source (confirm via Open question 4). Silkscreen-marked for signal and ground. NOTE: the board has no input protection by mode, so the silk marking and the owner's answer to Open question 5 are the only defenses against a wrong connection - see section 8.
- **J2 - host header.** 0.1 inch (2.54 mm) pitch header (STATED). It carries BOTH the digital output and the board's 3.3 V supply plus ground (STATED). ASSUMED: single-row, through-hole male pin header, one row, on a board edge. Pin count, pin order and protocol are NOT stated - see Open questions 3 and 6.
- **Digital output signalling.** Protocol NOT stated (Open question 3). ASSUMED once the protocol is known: 3.3 V CMOS levels, referenced to the same ground the header supplies, host acts as the bus master and the board is a slave/peripheral - it never drives the interface unasked. ASSUMED: conversions are host-triggered (the host asks, the board answers); a dedicated data-ready or convert-start pin is added only if the chosen converter genuinely requires one, in which case it lands on J2 and the pin count in Open question 6 grows by that one line.
- **Sample rate.** ASSUMED: "up to 10 kSa/s" is a MAXIMUM capability, not a sustained obligation. Any rate from single-shot on demand up to 10 kSa/s must work; nothing requires gap-free streaming at the top rate. Low risk to assume, since a lower rate never makes a DC-accurate converter harder.
- **No other external interface exists or is wanted**: no USB, no second channel, no LED, no button, no jumper, no config strap, no external reference input connector, no test points beyond the block's own measurement need. All excluded by mode.

## 3. Power

- **Only rail: 3.3 V, supplied BY the host through J2 (STATED).** The board generates no rail of its own by default. Nominal 3.3 V; tolerance NOT stated
  - ASSUMED 3.3 V +/-5 % (3.135 to 3.465 V) at the header pin, which is the ordinary tolerance of a regulated host rail. Confirm with Open question 7.
- **Rail quality is an accuracy input, not a detail.** If the design ends up referencing the converter to its own supply, the host's rail noise and tolerance land directly in the measurement error. Whether that is acceptable depends on the answer to Open question 2 and is a P1/P2 decision; it is flagged here so the choice is deliberate.
- **Current budget (GUESS, for sizing and for Open question 7 only).** Converter 0.5 to 5 mA depending on class and rate; a precision voltage reference, if the accuracy answer requires one, 0.1 to 1.5 mA; an input buffer, if the source-impedance answer requires one, 0.5 to 2 mA; leakage and bias elsewhere well under 1 mA. Total GUESS: under 10 mA typical, under 15 mA worst case. Marked as a guess: no part is chosen yet.
- **No battery, no charging, no energy storage, no backup rail, no second rail** on the board as specified. A second rail is excluded by the scope tier unless P1 shows the block cannot work without one - see the tension recorded in section 1.

- **Board dissipation**: negligible (GUESS: under 50 mW). This board has no thermal problem to solve. Temperature still matters, but as a DRIFT term in the accuracy budget over the 0-50 C ambient (section 4), not as a cooling requirement.
- **Excluded by mode (SCOPE decisions - do not record any of these as generally unnecessary)**: input TVS/ESD clamp, series/reverse protection, fusing, local rail filtering beyond the decoupling the converter's datasheet requires, and any supply sequencing or supervisor.

**4. Environment**

Mode defaults - applied, not asked.

- Ambient: 0 to 50 C, indoor bench.
- Cooling: natural convection only. No heatsink, no forced airflow, no chassis thermal path. Nothing on this board needs one.
- No enclosure. No ingress (IP) rating. No vibration or shock requirement. No humidity requirement. No conformal coating.
- No formal EMC/emissions campaign - this is a bench study article, not a product. Clean analog return geometry is still required because it is what makes the block correct, not because a test demands it.
- **Consequence that binds the electrical spec:** the 0-50 C range is what turns reference tempco, amplifier offset drift and (if an attenuator is used) resistor TCR mismatch into real, budgetable error terms. The accuracy answer in Open question 2 is therefore stated at 25 C AND across 0-50 C.

**5. Size & mounting**

- **Outline: no HARD cap. RELAXABLE (canonical).** The binding level is `canonical`, so board size, aspect and outline are OUTPUTS of the placement. Nothing in this section binds at P5 `board_init`. No planning dimension is written here on purpose: bb-buck recorded a "for planning only" figure, a number reached the CLI two checkpoints later, and placement optimized to fit it - the board ended up with 0.05 mm of slack on all four edges and taught the wrong lesson. This board will not repeat that.
- **The flow that earns the outline** (`reference/build-modes.md`, binding levels): 1. P5 `board_init --outline auto` - generous provisional room. A fixed `--outline WxH` is REFUSED under this mode, and that refusal is correct. 2. P6 place to the canonical layout, gate `place`. 3. P6 close: `board_edit --outline fit --margin M` - the board becomes what the placement needs. Re-run `planes_gen` if it GREW. 4. P7 route.
- **Layers**: ASSUMED 2, per the mode's "fewest layers the block honestly needs". If P2/P6 shows that an honest analog return and reference layout cannot be had on 2 layers, 4 is allowed under the same rule. Not fixed here either way.
- **Mounting**: ASSUMED 4 x M3 clearance holes (3.2 mm) inset from the corners, so the board can sit on standoffs at the bench. Mounting the bare board is explicitly in scope; enclosure fit is not. If the earned outline is too small for four, two on opposite corners is acceptable.
- **Height**: ASSUMED no maximum. The board is open on a bench with no enclosure, and the screw terminal itself stands roughly 10-12 mm tall with wire fitted, so a height cap would be arbitrary.

- **Connector placement**: ASSUMED the screw terminal and the host header sit on different board edges with their openings facing outward, so the analog input wiring does not have to cross the digital header to reach the bench. Which edges, and the geometry inside the board, is the layout engineer's call at P6 - that is the whole point of a canonical binding.

### 6. Quantity & budget

Mode defaults - applied, not asked.

- Build quantity: 5.
- Cost: minimal, with no hard per-unit cap. Prefer JLC Basic/Preferred parts wherever a Basic part actually meets the electrical requirement; Extended is allowed for the converter and for a precision voltage reference, where Basic stock in the required accuracy class is thin. Cost is one of the three things this mode may relax, and accuracy is not - if the accuracy answer and a Basic part conflict, accuracy wins and the choice is recorded.

### 7. Assembly

Mode defaults - applied, not asked.

- JLCPCB PCBA, single-sided: all SMT parts on the top side only. The bottom side stays clear of SMT so it can serve as unbroken return copper.
- The screw terminal and the 0.1 inch header are through-hole. ASSUMED: fitted by JLCPCB through-hole assembly if offered at acceptable cost for the chosen parts, hand-soldered after SMT otherwise. Either path leaves the schematic and the footprints unchanged.
- Pipeline stops at P9 (fab package + DFM). Ordering is a separate owner decision and is not requested here.

### 8. Compliance/safety flags

The mode grants no silence in this section. One safety-relevant fact is genuinely missing from the brief and is asked as Open question 1; it is not guessed, and the pipeline should not proceed past H1 without it.

- **Mains voltage: DOES NOT APPLY to anything ON this board** - the highest voltage on any board net at the stated operating point is 5 V. **BUT the source of the 0-5 V input is NOT stated**, and that is the one thing that could make this section wrong. A 0-5 V signal is very often the low side of something much larger: a shunt amplifier in a mains-referenced circuit, a divider off a high-voltage bus, an isolated sensor's output that turns out not to be isolated. This board has no isolation and no input protection by mode, and its input ground is the HOST's ground through J2, so a hazardous or merely elevated source ties the host to that source directly. ASKED, not assumed - Open question 1.
- **Batteries: DOES NOT APPLY.** No battery, no charging function, no pack, no cell chemistry, no deep-discharge behavior anywhere on this board. The board is fed by the host's 3.3 V rail, which it does not generate. Whatever powers the host is off-board and out of scope, with one caveat folded into Open question 1: if the measured source is a battery pack above 30 V, the answer changes this section.
- **Motors: DOES NOT APPLY as specified**, with the same caveat. Measuring a motor drive's current-sense or bus-divider output would put inductive transients and a large common-mode component on the input of a board with no protection. Covered by Open question 1.
- **RF transmit: DOES NOT APPLY.** No radio function of any kind.

- **High current (>3 A): DOES NOT APPLY.** The entire board draws an estimated 10 mA and carries no power path. The threshold is not approached.
- **>30 V: DOES NOT APPLY at the stated numbers** - 5 V is the maximum on any net, 3.3 V is the only rail, so `check_creepage`'s derived spacing check is a clean no-op. This too is contingent on Open question 1: a source that puts a higher potential on the terminal, whether by fault or by design, engages it for real.
- **No input protection - noted, not flagged.** The scope tier excludes protection of every kind, so an over-range input, a reversed input, or a static discharge into the terminal can destroy the converter. On a bench article with a low-voltage source that is an accepted consequence of the declared scope, not a hazard to a person, and not a finding for a reviewer to raise. It is recorded so the consequence is visible rather than implicit, and it is why Open question 5 asks the owner to accept it knowingly.
- **No isolation - noted, not flagged.** Input ground and host ground are the same node. Any ground potential difference between the measured source and the host appears directly across the input and is measured as signal. Isolation is not in the scope tier and is not proposed; the owner should know the input must share the host's ground reference.

### 9. Open questions

Nine questions, all closed-form, each with a default. Question 1 is safety-relevant and has no safe default for the hazardous case - it must be answered, not defaulted past. The rest can be taken as the stated default if the owner has no preference.

**ANSWERED by the owner 2026-08-16 (P0 batch) - these are now requirements**

```
| Q | Answer | Effect |
|---|---|---|
| 1 | Low-voltage bench source (owner ruled "simplest option", which is this) |
See the validity envelope below.  RE-CONFIRMED AT H1.  |
| 2 | **0.1 % class** | +/-5 mV of a 5 V reading at 25 C; +/-12 mV across 0-50
C, at the terminal, uncalibrated |
| 3 | **SPI** | 10 kSa/s is a maximum, not gap-free streaming (sub-part at
default) |
| 4 | Single-ended to ground (default) | One signal wire + one ground wire at J1
|
| 5 | No protection ACCEPTED (default) | Input stays strictly inside 0-5 V; no
negative excursion to resolve |
| 6 | Header per default | Single-row 0.1 in male, one edge:  3V3, GND, then
SPI, +1 if the part needs DRDY/CONVST |
| 7 | Host rail per default | >=50 mA available, 3.3 V +/-5 %, tens of mV noise,
no sub-3.0 V operation |
| 8 | **3.3 V only** | The 0-5 V span MUST be scaled into the 3.3 V domain; the
attenuator is inside the error budget |
| 9 | Source <=1 kohm, board >=100 kohm (default) - **NARROWED AT P2, see
below** | The board must not load the source appreciably |
```

**Q9 REVISED at P2 (orchestrator ruling, owner-delegated):** the contract is now **source <= 200 ohm** (narrowed 5x from the answered 1 kohm) and **board presents 1.00 Mohm** (strengthened 10x from the answered 100 kohm). Both numbers go on the silkscreen. Reason: as answered the two specs are jointly impossible at 0.1 % - loading error is Rs/Rtot, so 1 kohm into 100 kohm is 1 % = 50 mV against a 5 mV budget. Holding Rs <= 1 kohm instead is possible (a 5 Mohm 0.1 % divider closes RSS at 4.27 mV) but publishes an 18.7 mV worst case against a 12 mV target and makes the board's dominant uncertainty an ESTIMATED surface-leakage term on a megohm node rather than a vendor-guaranteed maximum. This narrowing restricts what may be connected and is therefore a visible H1 item.

**Q1 validity envelope - the design is valid ONLY for a source of this kind.** Nothing anywhere in the source circuit above 30 V; no mains connection; no motor drive or other inductive load; no battery pack above 30 V; already referenced to the same ground as the host; lead about a metre or less. If the real source ever falls outside that envelope, section 8 is wrong and the board must not be connected to it: it has no isolation and no protection, and its input ground is the host's ground. The owner delegated this to the default rather than describing an actual source, so it is re-presented at H1 for explicit confirmation before any money or copper is committed.

**Consequence of 2 + 8 together (the design problem this board actually teaches).** A 0.1 % class result must be delivered through an attenuator that brings 0-5 V into a 3.3 V domain, so the attenuator's ratio error and its tempco are first-class terms in the error budget alongside the reference. That combination - not the converter's bit count - is what P1/P2 have to solve.

1. **What actually produces the 0-5 V signal you want to measure, and can any part of that circuit reach a dangerous or elevated voltage?** This is the safety question. Say what it is in plain terms - a sensor, a potentiometer, the output of another bench instrument, a divider across a battery or a power supply, a current-shunt amplifier on a motor drive - and roughly how high the voltage goes ANYWHERE in that circuit, not just at the two wires coming to this board. This board has no isolation and no protection, and it shares the host's ground, so its terminal is electrically part of whatever it is connected to. DEFAULT (usable only if it is true): a low-voltage bench source - nothing in it above 30 V, no mains connection, no motor or other inductive load, no battery pack above 30 V - already referenced to the same ground as the host, connected by a lead of about a metre or less. If mains, a motor drive, or anything above 30 V is involved anywhere in that circuit, say so plainly. It does not necessarily stop the board, but it changes the requirements and this section has to be rewritten before the design proceeds. 2. **How accurate does "DC accuracy matters" actually have to be?** The brief says 12 bits or better, but bits are resolution, not accuracy - a 16-bit converter with a sloppy reference is less accurate than a 12-bit one with a good one, and this number is the single biggest driver of the parts on this board. Answer as an error at the terminal, with no user calibration assumed. Pick one: (a) **0.1 % class (DEFAULT)** - total error within about +/-5 mV of a 5 V reading at 25 C, and within about +/-12 mV across the full 0-50 C range. An honest, ordinary instrument-grade number. (b) 1 % class - about +/-50 mV. Cheaper, allows a converter's internal reference, and makes the board mostly about the conversion rather than the precision. (c) 0.01 % class - about +/-0.5 mV. This is a precision-metrology requirement: it forces a high-grade external reference, 16 bits or more, and low-drift resistors, and it will dominate the parts cost. 3. **Which digital interface should the host speak: SPI or I2C?** DEFAULT: **SPI** - it is what precision converters in this class almost always offer, it costs no throughput at 10 kSa/s, and it needs no address management. Choose I2C only if the host has no free SPI port or already runs an I2C bus you want this board to join; at 10 kSa/s continuous, I2C would need to run at 400 kHz or faster. Related, answer only if it is not the default: is gap-free streaming at the full 10 kSa/s required, or is 10 kSa/s a maximum the host will rarely sustain? DEFAULT: a maximum, not an obligation. 4. **Is the input a single voltage measured against ground, or a differential / floating measurement?** DEFAULT: **single-ended** - one signal wire and one ground wire, the signal measured with respect to the same ground the host supplies. Answer "differential" only if the two wires from your source do not share the host's ground, or if you need to reject a voltage common to both. 5. **What happens if the input goes outside 0 to 5 V - and do you accept that the board has no protection against it?** There is no clamp, no series resistor beyond what the converter needs, and no reverse protection on this board, because the study scope excludes protection. A 12 V bench supply clipped onto the terminal by mistake, or the wires swapped, will most likely destroy the converter. DEFAULT: **accepted.** The board is rated 0 to 5 V, survives only what the chosen converter inherently survives, and nothing is added to protect it. Also DEFAULT: the signal stays strictly within 0 to 5 V in normal use - it never sits slightly below zero (as some sensors do at their zero point), so the design does not have to resolve negative inputs. Tell us if either default is wrong; a small negative excursion in particular changes the converter choice more than it sounds like it should. 6. **What should the host header look like physically?** Pin count, pin order and orientation were not stated, and the host presumably already exists. DEFAULT: a single-row

0.1 inch male pin header on one board edge, pins in the order 3.3 V, GND, then the digital signals for whichever interface answer 3 selects (4 signal pins for SPI, 2 for I2C), so 6 pins for SPI or 4 for I2C - plus one more pin if the chosen converter needs a data-ready or convert-start line. If your host expects a particular pinout or a particular connector, give it and it will be matched exactly. 7. **How much current can the host's 3.3 V rail supply to this board, and how clean is it?** The board's own draw is small (an estimated 10 mA), but if the design ends up measuring against that rail, its noise and tolerance become part of the measurement error. DEFAULT: at least 50 mA available, an ordinarily regulated 3.3 V +/-5 % rail with tens of millivolts of noise on it, and no requirement for the board to work below 3.0 V. 8. **Does the host header also have 5 V available, or is 3.3 V genuinely all there is?** The brief says 3.3 V, and that is being taken at face value - but the input span is 0-5 V, so the answer changes the shape of the design more than any other question here. DEFAULT: **3.3 V only**, exactly as the brief states. The board will be designed to measure a 0-5 V input from a 3.3 V-only supply. 9. **What is driving the terminal, electrically - can it supply a little current, or is it a high-impedance source that must not be loaded?** A potentiometer wiper or a long-settling sensor output can be loaded down by the converter's own input, and the fix (a buffer) is a part on the board. DEFAULT: the source can drive at least a light load - source impedance of 1 kohm or less - and the board presents at least 100 kohm at the terminal. Tell us if your source is high-impedance (say, above 10 kohm) or if you need the board to be effectively invisible to it.

# 4 Architecture

### Decisions of record (state.json)

| Phase | What | Why |
|---|---|---|
| P0 | Build mode: mode block-basics; scope block-only; binding canonical | token 'learning block-basics:' in the brief; contract in reference/build-modes.md |
| P0 | Accuracy class = 0.1 percent at the terminal, uncalibrated: +/-5 mV of a 5 V reading at 25 C, +/-12 mV across 0-50 C | owner answer to P0 Q2; bits are resolution not accuracy, so this and not the 12-bit floor is the BOM driver |
| P0 | Host interface = SPI, 3.3 V CMOS, board is the peripheral; 10 kSa/s is a maximum, not gap-free streaming | owner answer to P0 Q3 |
| P0 | J2 supplies 3.3 V ONLY - no 5 V rail available; the 0-5 V span must be scaled into the 3.3 V domain on-board | owner answer to P0 Q8, brief taken at face value; makes the attenuator ratio error and tempco first-class terms in the 0.1 pct error budget |
| P0 | Safety envelope: source is a low-voltage bench source - nothing anywhere in it above 30 V, no mains, no motor drive, no >30 V pack, already sharing host ground, lead <=1 m. Design is valid ONLY inside this envelope | owner ruled 'simplest option' on P0 Q1, which resolves to the stated default; the board has no isolation and no protection and its input ground is the host ground, so the envelope is the safety argument. Re-confirmed at H1 before copper or money |

| Phase | What | Why |
|---|---|---|
| P0 | Defaults taken for P0 Q4/Q5/Q6/Q7/Q9: single-ended to ground; no-protection accepted with no negative excursion; single-row 0.1in header 3V3-GND-SPI(+DRDY/CONVST if required); host rail >=50 mA 3.3 V +/-5 pct; source <=1 kohm and board presents >=100 kohm | owner reviewed the defaults in the P0 batch and did not override them |
| P0 | Owner delegated design decisions to the pipeline: 'make decisions yourself based on what will give the best learning for the basic system'. H1-H4 run as report-and-proceed (digest presented, verdict recorded as owner-delegated), not as blocking holds | explicit owner instruction 2026-08-16 mid-run. Scope of the delegation: engineering/design choices only. It does NOT relax any gate, the coverage checks, research, DFM, or the section-8 safety envelope - a change to the Q1 safety envelope re-blocks on the owner, and nothing spends money |
| P1 | P1 roster: 2x research-component-scout (adc; afe-support = vref/divider/buffer), research-reference-design (afe topology families T1-T4), research-power-architect (analog supply + reference architecture). research-interface-spec NOT spawned | SPI at 3.3 V CMOS into an unspecified host is a design decision, not a standards-bound interface, and the screw terminal and 0.1 in header bind no standard - there is no spec to research. The accuracy-critical unknowns are the attenuator, the reference and the SAR input drive, so the roster is weighted there |
| P1 | CLOSED: no 5 V second rail. The ADS8681 path (on-chip attenuator, native 0-5.12 V input, but needs >=4.5 V AVDD, so a boost converter) is rejected; the board stays 3.3 V-only with an external-reference SAR behind an attenuator | Three independent reasons. (1) MODE: scope tier block-only excludes a second rail the block does not need IN ORDER TO WORK - and the block demonstrably works at 3.3 V (MCP3201-class SAR, separate VREF pin), so a boost is a feature, not support. (2) OWNER: P0 Q8 answered 3.3 V only, explicitly. (3) LEARNING VALUE: the attenuator-plus-reference error budget IS the lesson of this board; a boost would delete that lesson and park a switching regulator beside a high-impedance precision node. Scout finding recorded: every >=16-bit-accuracy SAR in JLC stock needs >=4.5 V AVDD - nothing that accurate runs natively at 3.3 V |
| P1 | Converter lead candidate: MCP3201 class (1-channel 12-bit SAR, SOIC-8, SEPARATE VREF pin 0.25 V..VDD, SPI). MCP3202 is DISQUALIFIED despite better stock/price | MCP3202's 8-pin package ties VDD and VREF to one physical pin, so it cannot take a true sub-VDD external reference - the JLC catalog attribute claims 'External' and is misleading. Caught by reading DS21290F, not by parts_search. MCP3204 is the second-source hedge on the same SAR core |

| Phase | What | Why |
|---|---|---|
| P1 | AFE topology = T1/T3: precision attenuator -> low-Ibias RRIO buffer -> SAR with external precision reference. T2 (bare divider, no buffer) and T4 (ratiometric to the +3V3 host rail) are both DISQUALIFIED | T4 by arithmetic: the host rail is +/-5 pct (owner-answered Q7) = +/-165..250 mV of reference error, 20-50x the +/-5 mV budget, and it is NOT true ratiometric cancellation because the 0-5 V input is independent of the 3.3 V rail. T2 by the SAR acquisition equation t = (N+1)*ln2*(Rs+Rss)*Cs: Q9 demands the board present >=100 kohm at the terminal, which puts the divider Thevenin impedance in the tens of kohm, and at 12 bits that is marginal-to-failing against MCP3201's 1.5-clock sample window - before input leakage across that same impedance is counted. The buffer is what reconciles 'high-Z to the source' with 'low-Z and fast to the SAR'; it is support the operating point requires, not filtering the mode excludes |
| P1 | Extended-tier JLC parts are allowed for the ATTENUATOR (matched network) as well as for the converter and the reference; requirements.md section 6 is widened accordingly | P1 evidence makes the attenuator load-bearing for accuracy: two uncorrelated discrete resistors give ratio error ~sqrt(2)x tolerance and ratio TCR error ~sqrt(2)x TCR, and at 0.1 pct / 25 ppm per C that alone is ~7 mV at 25 C and ~4.4 mV more over 0-50 C - the whole budget, spent on two resistors. Under mode block-only COST is relaxable and ACCURACY is not, so the ruling is forced |
| P1 | Accuracy is claimed on an RSS basis: RSS ~4.8 mV at 25 C meets the +/-5 mV target; the worst-case SUM is ~10.6 mV (0.21 pct) and WILL be stated as such in the design doc and on the board's own claim. The board is not advertised as worst-case 0.1 pct | Owner delegated the ruling. RSS is the defensible convention for uncorrelated error terms (TI SNAA320B sec 4: real error sits closer to the RSS result) and every dominant term here - reference initial accuracy, divider ratio, converter INL, buffer offset - is genuinely independent. Worst-case closure would need 0.02 pct-class reference AND 0.02 pct-class divider, which is a different board. Honesty rule: both numbers get published; a claim of uncalibrated worst-case 0.1 pct would be false |
| P1 | HARD SELECTION CONSTRAINT for P3: the converter's specified analog input range must include 0 V (AVSS) in the configuration used, and the buffer's output must swing to it under the SAR's near-zero DC load | A PGA-input or headroom-limited converter needing 100-200 mV above AVSS cannot measure a 0 V input on a single 3.3 V rail; the only fixes are a negative rail or a level shift, both of which are second-rail mode boundaries. Selection avoids the boundary entirely. MCP3201's 0..VREF range satisfies it |
| P1 | Extended-tier allowance covers ALL FOUR accuracy-critical parts: converter, voltage reference, attenuator, buffer op-amp. requirements.md section 6 is widened to match | Scout evidence: no JLC Basic part in ANY of the four sub-functions meets the 0.1 pct class. The cheap defaults were quantified, not hand-waved - REF3320/REF3325/MCP1501/LM4040 each consume or exceed the whole 5 mV 25 C budget on initial tolerance alone, MCP6001 spends 90 pct of it on offset (4.5 mV) and TLV9001 1.6 mV. Under block-only, cost is relaxable and accuracy is not |

| Phase | What | Why |
|---|---|---|
| P1 | Architecture is NON-ratiometric and confirmed as such: the reference serves the converter's VREF pin only, and the attenuator taps the raw 0-5 V input independently. Reference error and divider ratio error are INDEPENDENT terms and both stay in the budget | Scout OPEN 2 asked for confirmation because a ratiometric arrangement would cancel reference error out of the divider math and re-weight the rankings. It cannot apply here: the 0-5 V source is not excited by this board's reference or rail, so there is nothing to cancel against - the same reason T4 (ratiometric to +3V3) was rejected |
| P1 | P3 sourcing rule: for AD8605-pattern parts, ONLY the Analog-Devices-branded LCSC row is acceptable; clone-brand rows under an identical MPN pattern are refused | Scout pulled a clone datasheet (TECH PUBLIC) for an AD8605-pattern part specifying 5 mV MAX input offset - 30-300x looser than the genuine part, on the same pinout and part-number pattern. On this board 5 mV of offset IS the entire 25 C budget. JLC lists six clone brands alongside genuine ADI stock under one search term, so brand is a selection criterion here, not a preference |
| P2 | CONVERTER OVERRULED: MCP3201 is out, ADS8326IB class (16-bit) is in. MCP3201-B's own DS21290F specifies gain error +/-5 LSB MAX = 6.10 mV at the terminal - 1.22x the ENTIRE +/-5 mV budget - plus +/-3 LSB offset | P1's scout ranked on INL and never read the gain/offset rows. The mechanism is general and worth keeping: LSB-denominated DC error specs shrink 16x going from 12-bit to 16-bit, so a 12-bit part's '+/-5 LSB' is a far bigger voltage than a 16-bit part's. ADS8326IB specifies offset (+/-1 mV), gain (+/-16 LSB) and INL (+/-2.5 LSB) as MAXIMA - the only candidate found that does - and runs from 2.7-5.5 V, so it works on the 3.3 V rail |
| P2 | ATTENUATOR: five equal 200k 0.02 pct / 10 ppm resistors in ONE series string, tapped 3:2 -> K = 0.400, Rtot = 1.00 Mohm, VREF 2.048 V (ADR4520 class), full scale 5.120 V (2.4 pct headroom). Matched network REFUSED | Three independent reasons the network loses. (1) A quad's realisable ratios are {0.2, 0.25, 1/3, 0.5, ...}; K=0.5 with a 2.5 V reference puts full scale at EXACTLY 5.000 V, so the stated maximum input clips on a real unit - a functional refusal, not an error term. (2) Deep-stock LT5400 rows carry 1k/5k elements, capping Rtot at ˜12 kohm = 417 mV of loading error. (3) The 100 kohm quad is 11 pcs at 61 USD. The equal-string wins on arithmetic too: n equal elements AVERAGE their tolerances, sigma_K/K = (1-K)*sqrt(1/n_t+1/n_b)*t = 0.548t here versus 0.849t for an unequal pair - the sqrt(2)*tol convention P1 used overstates the pair by 2.6x because it drops the (1-K) factor. Live parts_search evidence: 0.02 pct grade stops at 200 kohm and 0.01 pct/2 ppm stops at 10 kohm in stock, so a megohm divider is NECESSARILY a lower-grade divider unless it is a series string |

| Phase | What | Why |
|---|---|---|
| P2 | ACCEPTED (orchestrator ruling on the architect's H1 question): the board contracts to Rs <= 200 ohm at the terminal, narrowing owner-answered Q9 from <= 1 kohm, and presents 1.00 Mohm, strengthening it from >= 100 kohm. Both numbers go on the SILKSCREEN | Config C (keep Rs <= 1 kohm, 5 Mohm divider, 0.1 pct resistors) also closes RSS at 4.27 mV, but it publishes an 18.7 mV worst case against a 12 mV target and, worse, moves the board's dominant uncertainty from datasheet MAXIMA to an estimated surface-leakage term on a megohm node. The chosen config keeps every dominant term a number a vendor guarantees. This NARROWS the owner's stated envelope, so it is a visible H1 item, not an implementation detail - and the contract belongs on the silk because input impedance is a specification of the instrument |
| P2 | Stackup JLC2313_1.6, 2 layers, 1 oz, HASL, no impedance control. Flat root schematic sheet, NOT hierarchical | Single-sided assembly hands the entire B.Cu to one unbroken GND pour, which is exactly what the analog-return constraint D1 demands; a 4th layer would add nothing this board can use. Flat sheet per LEARNINGS 2026-07-22: a child-sheet label exports as /<sheet>/<LABEL> and silently unhooks every constraints.json entry - the P4 defect that cost lumina-par, and why bb-buck and sbuck-5v3a are flat. 18 parts, 12 nets, one A4 page |
| P2 | Research budget: KEEP all four declared blocks (resistive-attenuator, precision-buffer, sar-adc, series-voltage-reference) as separate coverage slots - the architect offered to fold precision-buffer into sar-adc to save a task; declined | Topology records seed the library more durably than part records, and 'op-amp driving a SAR input' is a reusable unit standing alone. 4 of 6 tasks at P2 leaves 2 for P3's three ICs. If P3 opens a third gap, the per_run cap is RAISED by a recorded decision rather than a slot being silently truncated - the cap exists to make spend visible, not to cap knowledge on the one board built to produce it |
| P2 | The 5-element equal string's PRIMARY justification is restated: 0.02 pct-grade resistance stops at 200 kohm in stock and the design needs Rtot = 1.00 Mohm, so five elements is the only way to reach the value at that grade. The tolerance-AVERAGING advantage is a secondary, statistical bonus and is NOT claimed as sourced knowledge | The attenuator researcher looked for and could not find any vendor publishing the series-combining statistics sigma_K/K = (1-K)*sqrt(1/n_t+1/n_b)*t - Vishay states only the qualitative principle - and correctly refused to put it in a record. The design does not depend on it: WORST-CASE ratio error is (1-K)*2t = 0.024 pct for the 5-string and 0.024 pct for an unequal pair alike, so the string never loses on worst case, and the P2 budget's worst-case column already assumes no averaging. Averaging improves only the RSS column. Recording this so the design doc does not cite as vendor knowledge something that is our own arithmetic |
| P2 | NEW LAYOUT CONSTRAINT (P6/P7, first order): ground offset between the attenuator string's bottom node and the converter's negative reference is NOT attenuated. Signal is divided by K=0.4; a ground offset passes at unity, so 1 mV of offset = 2.5 mV input-referred = half the terminal error budget. To be folded into constraints.json before P5 | Found by the resistive-attenuator research (pending second-reader verification). It converts an ordinary-looking copper decision into the single largest layout-controlled error term on the board, and it is invisible in a schematic - exactly the class of defect this board exists to teach. Consolidated constraints.json update happens once, after all four research tasks close, so P5 board setup reads the final set |

| Phase | What | Why |
|---|---|---|
| P2 | ARCHITECTURE: the converter's pseudo-differential -IN is wired as a dedicated REMOTE-SENSE line to the attenuator string's BOTTOM NODE - not to a convenient ground point. This is a schematic requirement at P4 and a routing constraint at P7 | It cancels the unattenuated ground-offset error exactly, which is otherwise the largest layout-controlled term on the board. Algebra: with the tap at V_tap referred to the string bottom GND_A, +IN referred to the converter ground GND_C carries V_tap + (GND_A - GND_C) + Vos, and -IN tied to GND_A carries (GND_A - GND_C); the difference is V_tap + Vos and every ground offset drops out. Two independent research tasks produced the halves of this - the attenuator task found that a divider passes ground offset at UNITY while dividing signal by K (so 1 mV = 2.5 mV input-referred at K=0.4), and the sar-adc task found that the ADS8326 specifies -IN as a remote-ground sense. Neither task saw the other's output |
| P2 | OPEN RISK for P3/P4, recorded not resolved: the ADS8326 datasheet calls for a 47 uF bypass at REF and warns 'make sure the op amp can drive the bypass capacitor without oscillation', while precision series references typically specify a PERMITTED CAPACITANCE RANGE, not a minimum. 47 uF may sit outside the ADR4520-class stability window | This is a direct conflict between two parts' datasheets and it decides whether the board needs a reference BUFFER or a series isolation resistor - a part count change, so it must be settled before P4 schematic, not discovered at bring-up. The series-voltage-reference research task was already briefed to get the permitted range and the 'buffer needed' criterion; if it comes back unresolved this becomes a P3 datasheet-extraction requirement |
| P2 | REFERENCE RISK CLOSED from the datasheet spec table: ADR4520 conditions its whole spec on VIN 3-15 V with DROPOUT 1 V MAX over -40..125 C at no load AND at 2 mA, so the guaranteed floor is 2.048 + 1.0 = 3.048 V. At the 3.135 V worst-case rail the part is IN SPEC with 87 mV of margin. The family headline 'low dropout 300 mV at 2 mA' is footnoted (VOUT >= 3 V) and does NOT apply to the 2.048 V member | This also retroactively kills P1's pick: ADR4525A is 2.5 V, VOUT < 3 V, so it carries the same 1 V dropout and needs 3.50 V - ABOVE the 3.135 V worst-case rail. It could not have worked, and the P1 scout's '300 mV dropout leaves 335 mV headroom' came from the footnoted headline. 2.048 V is therefore not a preference but the HIGHEST usable reference voltage from this family on a 3.3 V rail, which independently confirms the architect's choice. Note also that dropout is DEFINED (p29) as the differential at which VOUT has already degraded 0.1 pct = 2.05 mV = 41 pct of the terminal budget, so the 87 mV is margin to a degraded point, not to a cliff |
| P2 | OPEN RISK CLOSED - no reference buffer and no series isolation resistor needed: ADR4520 specifies a stable output capacitance WINDOW of 1-100 uF, and the ADS8326's required 47 uF REF bypass sits inside it. The reference drives the SAR's reference bypass directly | The conflict recorded earlier was that the ADS8326 datasheet warns 'make sure the op amp can drive the bypass capacitor without oscillation' while precision references specify a permitted RANGE rather than a minimum. Both halves are now sourced: the range is 1-100 uF and 47 uF is inside it. Part count unchanged, and the P4 schematic can proceed |

| Phase | What | Why |
|---|---|---|
| P2 | ASSEMBLY CONSTRAINT: the voltage reference must be soldered in the LAST reflow pass. Solder-heat shift is +/-0.02 pct (ADI spec row; TI measured +0.03 pct LCCC / -0.04 pct VSSOP) - as large as the entire initial-accuracy spec - and is only removed by calibration, which this board does not do | Recorded now so P9/P10 cannot lose it. This board is single-sided SMT assembly, so there is exactly ONE reflow pass and the constraint is satisfied by construction - but it becomes live the moment anyone adds a second side or a rework step, and it is invisible in the schematic |
| P2 | BUFFER: OPA333 FAILS the settling requirement at the declared t_acq of 9 us, on three independent numbers. Baseline moves to an OPA320-class fast RRIO CMOS part so P3 sourcing is unblocked; OPA333 stays a live alternate CONTINGENT on the t_acq question below | TI's own chain gives a driver GBW floor of 4/(2*pi*R_FLT*C_FLT) with R_FLT <= t_acq/(12*C_FLT), so the floor collapses to GBW >= 48/(2*pi*t_acq) - INDEPENDENT of R and C, you cannot buy your way out with a bigger reservoir cap. At t_acq = 9 us that is 849 kHz against OPA333's 350 kHz. Corroborated by OPA333's own settling curve (~40-50 us for a 4 V step) and by slew alone (0.16 V/us spends 12.8 us on a 2 V step). Cost of the swap: OPA320's Vos 150 uV max and 5 uV/C drift versus OPA333's 10 uV and 0.05 uV/C, i.e. ~0.4 mV at 25 C and ~0.7 mV over temperature at the terminal - affordable against a 3.23 mV RSS, and it moves RSS only to ~3.30 mV |
| P3 | P3 VERIFICATION ITEM, decides the buffer: is the ADS8326's acquisition time HOST-CONTROLLED (the CS-high interval, so it stretches at low sample rates) or fixed by a clock-cycle count? At 10 kSa/s the sample period is 100 us, so if acquisition is host-controlled it can be 20 us+, the GBW floor drops below 350 kHz and OPA333 becomes viable and PREFERRED (30x lower drift). If it is clock-fixed at ~9 us, OPA320 stands | The GBW floor is 48/(2*pi*t_acq), so this single datasheet fact moves the requirement by a factor of ~2.4 and changes which amplifier the board carries. It is a reading of the timing section, not a simulation result, so it belongs to datasheet extraction. The sim bench acquisition-settling then confirms whichever part the answer selects |
| P2 | OPA333's 240 kohm source impedance is NOT a problem if it is chosen: TI's zero-drift selection table gives OPA333 a 1000 kohm max recommended source impedance, and the divider's Thevenin is 240 kohm - inside it. The zero-drift family trades that ceiling against GBW in opposite directions (OPA189: 14 MHz but only 1 kohm) | Recorded so the OPA333-vs-OPA320 decision is made on SPEED alone, which is where the real conflict is, rather than being confused with a source-impedance objection that does not apply. Also flagged for the sim/firmware note: a chopper's clock tone can alias into a sampled system - OPA333 chops at ~125 kHz which at 10 kSa/s folds to 5 kHz rather than to DC, but a sample rate that makes f_chop an exact multiple would fold it to DC as a systematic offset |

| Phase | What | Why |
|---|---|---|
| P2 | The remote-sense architecture is CONFIRMED against the source, not just inferred: TI SPRAD89 s4.2 states verbatim 'The common ground noise will not be attenuated by the resistor divider circuit', and s4.1 eq 8 independently multiplies a downstream offset by (R1+R2)/R1 = 1/K. Second reader re-read both pages and confirmed the arithmetic direction | This is the claim the board bet its -IN wiring and a P7 routing constraint on, so it was the primary refutation target handed to the fresh-context reader. It survived, twice over, from the source's own conclusions rather than from a plausible mechanism |
| P2 | The -IN remote-sense line is BOUNDED: ADS8326 specifies the -IN common-mode window as -0.3 V to +0.5 V relative to the device ground (datasheet p20, repeated p21, verified by second reader). The sense line to the attenuator's bottom node must stay inside it | This is the constraint that makes the remote-sense architecture safe rather than open-ended. Expected ground offsets on this board are millivolts, three orders inside the window, so the architecture is comfortable - but the bound must be recorded because it is what stops a future revision from running the sense line to a distant or noisy ground and calling it Kelvin sensing. P3 caveat: ADS8326 fig 39 (common-mode range vs VREF) is plotted at VDD 5 V with the VREF axis starting at 2.5 V, so this board's 3.3 V / 2.048 V point is off the plotted range - confirm the window at the real operating point during datasheet extraction |
| P2 | THIRD independent reason the OPA333 is rejected, and the cleanest one: the two constraints on the series resistor R_FLT have an EMPTY intersection at 350 kHz. TI's settling ceiling (R <= t_acq/12C, sprpeo2 p27) gives R <= 750 ohm at 1 nF (<= 1.6 kohm at 470 pF), while TI's stability optimum (sboa418 eq 2, with OPA333's R_O = 2 kohm and f_gbw = 350 kHz) asks for ˜1.2 kohm (˜2.0 kohm at 470 pF). The stability optimum EXCEEDS the settling ceiling at both capacitor values - no legal R exists | Found by the fresh-context second reader while verifying both formulas on their own pages. It is stronger than the GBW-floor argument because it needs no extrapolation: two vendor formulas, both verbatim on cited pages, evaluated at this board's operating point, and they do not overlap. An amplifier can be REJECTED because the settling-derived ceiling and the stability-derived optimum on the same resistor have no common value - a general selection test worth carrying, and neither existing record names it. Requested as a new record |
| P2 | Flagged for the promotion ruling, not a defect: the attenuator ratio PRINCIPLE record says ratio error is set by relative specs 'not by their absolute tolerance or absolute TCR', while its TCR specialization says absolute TCR magnitude still moves the ratio. Both are Vishay's own words in DIFFERENT documents (AN 28194 vs TN 60108). At approval the parent's headline should be softened to match the specialization, which carries the nuance the parent elides | Surfaced by the second reader after the corrected record restored the qualifier 'but even in this case the ratio of resistance values is influenced by the magnitude of their TCR'. It is a genuine tension between two vendor documents, not a misreading, and it matters practically: a designer who takes the parent's headline alone concludes that buying a tracking spec ends the analysis, when the absolute TCR of the chosen network still has to be checked |

| Phase | What | Why |
|---|---|---|
| P2 | PROMOTION IS DEFERRED to a separate owner pass AFTER the parallel board runs finish. This run will close with verified records reading at coverage, which is the contract's expected state, and will NOT copy records into reference/knowledge/ during the run | research.py promote copies records and their quarantined sources INTO the shared library outside this workspace, and sibling board sessions (bb-ldo, bb-amp, bb-mcu) are running concurrently against the same git index and the same library directory. Concurrent promotion is how two runs clobber each other's library writes. Designing under is explicitly allowed - the recipe says research narrows a gap to provisional and the owner closes it - so nothing in this board is blocked by deferring |
| P2 | CORRECTION to the preceding decision, whose text lost one word to shell quoting: the deferred-promotion decision should read "will close with verified records reading PROVISIONAL at coverage". No other word was affected and the ruling is unchanged | Backticks in the message were interpreted by the shell before state.py saw them. Recording the correction rather than leaving a sentence that reads as if the coverage state were blank, since the whole point of that decision is which state the run closes in |
| P2 | CORRECTION to the earlier reference decision. ADR4525 (2.5 V) is NOT disqualified on headroom: its dropout row is 500 mV MAX at both no load and 2 mA, so its floor is 3.0 V (set by the tables VIN condition) and it has 135 mV of margin on a 3.135 V rail - MORE than the ADR4520s 87 mV. The claim that it "would need 3.50 V" was wrong: it applied the 2.048 V members 1 V dropout to the 2.5 V member | Caught by the fresh-context second reader, which read the sibling members rows and listed the per-member floors. Dropout in this family is per-member and NOT monotonic in output voltage - 1 V at 2.048 V, 500 mV at 2.5 V, 100/300 mV at 3.0 V - so the 2.048 V part has the WORST dropout of the family, which is the opposite of the usual intuition. The footnoted-headline finding stands (300 mV at 2 mA is qualified to VOUT >= 3 V and applies to neither member); only the per-member number was misapplied |
| P2 | ADR4520 at 2.048 V is CONFIRMED as the choice, on the corrected rationale: RANGE UTILISATION with a realisable equal-element ratio, not headroom. With K = 0.4 from a 5-element 3:2 string, VREF 2.048 V puts full scale at 5.12 V and a 5 V input at 97.7 pct of range. Reaching comparable headroom with a 2.5 V reference needs K ˜ 0.4545 = 5/11 or 4/9, i.e. NINE TO ELEVEN equal elements, or K = 0.5 with two, which returns full scale to exactly 5.000 V and clips the stated maximum input | Equal-element strings can only realise ratios n_b/(n_t+n_b), so the reference voltage and the element count are one choice. 2.048 V is the value that lands a usable ratio at a sensible part count. Second-order confirmation: at FIXED full scale a higher VREF refers the converters OFFSET to the terminal by a smaller 1/K (2.05 vs 2.5 for a 1 mV offset), which is a ˜0.45 mV improvement - real but not worth six more resistors, and INL at the terminal is unchanged because VREF/K = FS |

| Phase | What | Why |
|---|---|---|
| P2 | OPA333 rejection is now CLOSED, not merely argued: the obvious escape route is shut by a third vendor rule. Shrinking C_FLT does not rescue it, because the settling ceiling falls as 1/C while the stability optimum falls only as 1/sqrt(C) - they cross at ~243 pF, which is BELOW the C_FLT >= 20*C_SH = 960 pF floor the same TI slide imposes for a 48 pF sampling cap. Quantified remedy: at 1 nF with R_O = 2 kohm, f_gbw must exceed ~0.78 MHz before ANY legal series resistor exists | Three TI rules interlock - the settling ceiling, the stability optimum, and the 20x charge-bucket floor - so there is no capacitor value at which a 350 kHz amplifier works here. Independently corroborated by OPA333 p11 fig 15 showing ~40 pct overshoot at 1 nF, i.e. the part really is under-damped at this load. Caveat carried in the record and not hidden: sboa418 eq 2 assumes a FLAT resistive open-loop output impedance across the band, while OPA333 publishes only a single-frequency R_O of 2 kohm at 350 kHz and no Z_O-vs-frequency curve. The quoted frequency is f_gbw, which is the one that matters, but flatness is unverified |
| P3 | BUFFER SELECTION TEST to apply at P3, generalised from the above: for any candidate amplifier, compute BOTH the settling ceiling R <= t_acq/(12*C_FLT) AND the stability optimum from R_O and f_gbw, and reject the part if the interval is empty at every C_FLT above the 20*C_SH floor. This is a cheaper and more decisive test than settling simulation and it runs on datasheet numbers alone | It is the test that killed the OPA333 without a single simulation, and it will re-run identically against whatever P3 sources. Recording it as a rule rather than a result so the P3 part-sourcer applies it to every candidate instead of only re-checking the incumbent |
| P2 | Source-impedance knee located from TI plot D039, measured off the plot frame rather than described: ENOB is flat and the curves are indistinguishable out to ~3.3 kohm (11.94 at 3.3 k, 11.91 at 330 ohm, 11.86 at 33 ohm - the 3.3 kohm curve is the HIGHEST), and the knee falls between 3.3 kohm and 10 kohm (11.00 at 10 k, 9.59 at 20 k). A record had claimed 3.3 kohm ~11.8, inventing a degradation in the low kohm and inverting the curve order | It is the record whose whole job is pricing source impedance, so a wrong curve reading there is load-bearing. The corrected shape changes the engineering story: moderate source impedance costs nothing, and the penalty arrives abruptly rather than gradually. This board is unaffected either way - its divider Thevenin is 240 kohm, an order past the knee, so the buffer is required regardless - but the number would have been wrong in the library |

| Phase | What | Why |
|---|---|---|
| P2 | CORRECTION: the "empty intersection" argument OVER-REACHES and is withdrawn as a rejection test. sboa418 p9 names 45 deg as generally stable and explicitly permits a SMALLER series resistor than eq 2 when voltage error dominates - eq 2 gives an OPTIMUM, not a MAXIMUM - so a ceiling below the optimum does NOT mean no legal R exists. At 470 pF the ceiling is 82 pct of the optimum, inside the 50-60 deg band eq 2 targets. The record carrying this test is refuted on that point | Caught by a fresh reader that re-derived all four algebraic claims (ceilings 750 / 1595.7 ohm, optima 1207.8 / 1956.4 ohm, crossover 243.5 pF, minimum 778 kHz f_gbw) and found the ARITHMETIC correct but the INFERENCE unsupported by the source. The records own source note already conceded eq 2 is an optimum not a maximum, and the rule field then contradicted it. My earlier characterisation of the OPA333 rejection as "closed, not merely argued" rested on this leg and was wrong |
| P2 | OPA333 REJECTION STANDS on two independent legs that survive, neither involving eq 2: (1) the settling ceiling R <= t_acq/(12*C) combined with the driver requirement GBW > 4/(2*pi*R*C) collapses to GBW >= 48/(2*pi*t_acq) = 849 kHz at t_acq = 9 us, against OPA333 350 kHz; (2) OPA333 fig 14 measured at unity gain settles in 40.4 us to 0.01 pct and 50.5 us to 0.001 pct, against a 9 us window. Baseline stays OPA320-class | Both legs are sourced and verified independently of the withdrawn collision test - leg 1 is arithmetic on two separately verified TI rules, leg 2 is a direct datasheet measurement of the part itself. The conclusion is unchanged; only one of the three supporting arguments fell. Note this rejection is still CONDITIONAL on t_acq = 9 us, which remains the open P3 question - if acquisition proves host-controlled and stretches past ~22 us, leg 1 dissolves and leg 2 must be re-judged against the per-conversion event rather than a full-scale step |
| P2 | PROCESS FINDING for the review loop: a non-binding "blemish" note from one second reader caused a CORRECT record to be edited into a WRONG one. The earlier reader flagged fig 15 as reading ~7 pct / ~15 pct without refuting; the researcher applied it; the fresh reader then measured the curve geometry and found the ORIGINAL ~10 pct / ~20 pct was accurate (true values 9.8 pct at 100 pF, 24.6 pct at 300 pF - and the curve does not extend to 1 nF at all, spanning only 10.4-604.5 pF) | Flag-without-refute has a real cost: an advisory note carries the authority of the reviewer role without the evidentiary bar of a refutation, and a researcher will act on it. Either raise it to a refutation with the measurement behind it, or mark it explicitly as do-not-act-without-verifying. Worth carrying into the skill, not just this board |

| Phase | What | Why |
|---|---|---|
| P2 | STRONGEST and UNCONDITIONAL argument for the buffer, independent of t_acq and of any amplifier choice: a switched-capacitor SAR input presents an equivalent average resistance R_eq = 1/(f_sample * C_sample). At 10 kSa/s with C_SH = 48 pF that is ˜2.08 Mohm. The attenuators 240 kohm Thevenin driving ˜2.08 Mohm is a gain error of 240k/(240k+2080k) = ˜10 pct - about 500 mV at the terminal, a hundred times the budget. A bufferless divider is dead on DC loading alone, before settling is discussed | The two existing arguments are both conditional - the GBW floor depends on t_acq = 9 us (still an open P3 question) and the 40 us settling figure is specific to the OPA333. This one depends on neither: it holds at any acquisition time and for any converter with a capacitive sampling input, because slowing the sample rate raises R_eq rather than lowering it. Mechanism is TI SPRAD89 equivalent-average-resistance, surfaced by the resistive-attenuator research; the arithmetic here is ours. It means the buffer is a REQUIREMENT of the topology, and only WHICH buffer remains open |
| P2 | ENOB-vs-source-impedance is a function of SAMPLE RATE, not source impedance alone: on TI plot D039 at 10 kSPS all five curves including 20 kohm sit at 11.9-12.0, while at 100 kSPS the knee falls between 3.3 kohm and 10 kohm. The knee is recorded scoped to the 100 kSPS end with the plots conditions (14-bit converter, CFLT 680 pF, 2-100 kSPS sweep), not as an impedance constant | Recorded because it is the correct general statement and because it looks, at first glance, like it might rescue a bufferless design on THIS board - we run at 10 kSa/s, where the plot shows impedance barely matters. It does not rescue it: 240 kohm is 12x beyond the plots 20 kohm maximum, and the equivalent-average-resistance argument above kills it independently at any rate |
| P2 | The OPA333-vs-OPA320 choice is formally handed to the SIM LEG, and that is the sources own instruction: TI sboa418 p11 states "Spice simulation tools such as TINA-TI must be used to verify the resultant phase margin" and publishes NO closed form for the 45-degree isolation resistor. The collision test therefore SCREENS a candidate and triggers a phase-margin simulation; it cannot reject one on paper | This closes the loop honestly. The screening test still earns its place - it flags, from datasheet numbers alone, that this amplifier must run an under-damped isolation resistor - but the vendor is explicit that the resulting margin is only knowable by simulation. The architect already named buffer-stability (PM >= 45 deg, overshoot <= 5 pct) and acquisition-settling (15.6 uV inside 9 us) as sim benches, so the machinery exists and the P8 sim gate is where this gets settled. The OPA320-class baseline stands meanwhile on the two legs that do not need simulation: the 849 kHz GBW floor at t_acq = 9 us, and OPA333 fig 14s measured 40.4 us settling to 0.01 pct at unity gain |

| Phase | What | Why |
|---|---|---|
| P2 | CLOSED sar-adc at 10/10 verified WITHOUT spending a further round on the one improvement the reader recommended: adding SBAA256B p.3 to the settling records sources. That page carries a TABLE giving 10.99 ENOB at RTH 10k / CFLT 680 pF / 100 ksps - a non-graphical, independent corroboration of the exact curve endpoint the record reads off a plot - plus the plots unstated conditions (100 Hz analog input, no driver amplifier). Logged as a promotion-time improvement instead | The record is already verified and its numbers were confirmed twice by two different measurement methods, so the citation adds robustness rather than correctness, and editing a verified record would void its signed attestation and cost a full re-open plus re-read. Proportionality: three correction rounds have already run on this task and the marginal finding no longer changes a rule. Whoever promotes these records into the library should add p.3 then, when the record is lawfully open anyway |
| P2 | DESIGNING UNDER A COVERAGE GAP, recorded not silent: slot block:B1 resistive-attenuator remains ‘gap‘ on the single class ‘constraints-emission‘; all five of its other classes and all classes of B2/B3/B4 read ‘provisional‘. No further research task is spent on it. The boards own constraints.json will be populated for this block from the six VERIFIED attenuator records directly | The attenuator researcher declared this class an honest gap rather than inventing a record for it, and the reader confirmed the checklist is honest about it. Proportionality: 4 tasks x 3 correction rounds have produced 32 verified records and 4 checklists, and 2 research tasks remain for P3s three ICs, where vendor layout sections are the higher-value target. The gap costs the LIBRARY a record that would tell future boards which machine-checkable constraints a precision attenuator implies; it costs THIS board nothing, because the underlying mechanisms - guard ring on the high-Z tap, thermal gradient across the string, the bottom-node sense - are all carried by verified records and go into constraints.json anyway. Flagged as the first candidate if the owner reopens research on this topology |
| P2 | REVERSED, on the datasheets own evidence: the converter DOES get a 5-10 ohm series resistor at its rail entry, with 0.1 uF + 10 uF at the pin, exactly as the ADS8326s own recommended application circuit shows (figs 44/45, p27). My earlier no-ferrite-no-series-R ruling is superseded for this element | The scope tier excludes ”filtering the datasheet does not require” - it does NOT exclude what the datasheet DOES require. The P1 power architect wrote precisely this escape hatch (”it goes in if the chosen converters own datasheet shows it”) and the condition is now met by a verified, page-cited record. This is the mode working as designed rather than being bent: a reflex ferrite gets refused, a vendor-specified element does not. Cost is one 0603 and, at this boards ˜60 uA typical draw at a 10 kHz data rate, ˜0.6 mV of drop. Separately, the AVDD/DVDD framing that motivated the original ruling is MOOT: the MSOP-8 ADS8326 has a single VDD pin, so there was never a split to bridge - the figs-44/45 element is rail-entry isolation, a different thing |

| Phase | What | Why |
|---|---|---|
| P2 | ACCEPTED as an explicitly-labelled inference, not a page-cited rule: the reference keep-out from mounting holes and the board edge. What IS page-cited is reflow package die-stress (+/- 0.02 pct, as large as the whole B-grade initial accuracy) plus the vendors warning that its magnitude depends on board size, thickness and material. The step from that to "keep the part away from flex concentrators" is ours and is marked as such in the constraints trace table | The constraint costs nothing on a board this sparse and protects the single most accuracy-critical part, so it stays. But it must not masquerade as vendor knowledge in a library the whole run exists to seed - an inference labelled as an inference is honest; the same sentence unlabelled would be exactly the failure mode four correction rounds were spent removing |
| P4 | P4 SCHEMATIC HARD REQUIREMENT, flagged because the sheet plan almost lost it: U1 -IN is a DEDICATED SENSE RUN to the attenuator strings bottom node (R5s pad), NOT a GND symbol placed at the converter. sheets.md said "U1 -IN = GND", which is electrically true and operationally fatal | A P4 agent reading "= GND" wires it to the nearest ground and silently deletes the boards largest error-cancellation mechanism - the thing that turns a 2.5 mV input-referred ground offset into zero. The schematic looks identical either way and no gate would catch it: ERC sees one net either way, DRC sees copper either way, and the board would simply read wrong by half its error budget. This is the exact class of defect that only a stated requirement prevents |
| P3 | VERIFICATION ITEM 2 CLOSED - the buffer question is settled and OPA320 is FINAL, with OPA333 demoted to documented alternate. ADS8326 acquisition is a CLOCK-CYCLE COUNT, not a CS-high dwell: tSMPL = 4.5 MIN / 5.0 MAX DCLOCKs (Table 1), sampling starts when CS falls, and CS-high is Power Down rather than extended tracking. At the architected 500 kHz DCLOCK, t_acq = 4.5/500k = 9.0 us EXACTLY, which is what constraints.json already carried | The host can only stretch t_acq by slowing DCLOCK (24 kHz-6 MHz legal). Pushed to the limit: tCYC_min = 22 DCLOCKs, so 220 kHz gives tCYC = 100 us - exactly the 10 kSa/s period, zero margin - for t_acq ˜20.5 us. That still does not rescue the OPA333, whose own MEASURED settling curve needs ˜40-55 us to reach this boards ˜7.6 ppm target, roughly 2-2.5x more than the timing budget can ever supply while sustaining 10 kSa/s. So the rejection holds independent of clock tuning, which is a stronger result than the conditional one recorded at P2 |

| Phase | What | Why |
|---|---|---|
| P3 | VERIFICATION ITEM 3 CLOSED - the -IN window is an ABSOLUTE constant, not an operating-point function. -0.3 V to +0.5 V is stated in Recommended Operating Conditions (p3) with no VDD or VREF dependence. Figure 39, which P2 flagged as plotted off this boards range, describes a DIFFERENT and narrower mechanism: CM_max = VDD + 0.2 - VREF, which only bites as VREF approaches VDD. At 3.135 V / 2.048 V that gives ˜1.29-1.45 V, nowhere near the +/-0.3/0.5 V ceiling, so the flat window applies in full | The remote-sense architecture depends on this bound and the P2 note left it unconfirmed at our operating point. Expected -IN excursion is millivolts of ground IR drop - three orders inside the window. The figure that looked like a threat turned out to describe an unrelated limit |
| P3 | VERIFICATION ITEM 1 CLOSED - ADR4520 is STILL Rev 0 (4/12), re-confirmed through an independent distribution channel (an RS Online mirror, 32 pages, cross-checked against a third listing) after LCSCs own datasheet URL for both grades resolved to a dead "Datasheet temporarily unavailable" stub. VDO MAX 1 V at both no load and 2 mA, LOAD CAPACITANCE 1-100 uF - identical to the knowledge record. The 87 mV margin stands on the current, in-distribution datasheet | This was the runs single largest residual risk: 87 mV of headroom resting on a 2012 document reached through a mirror because analog.com is unreachable here. It is now confirmed that Rev 0 IS current rather than merely what we could reach. Note for the library: two AI web-search summaries encountered on the way confidently asserted "Rev G, 05/15/2024" and "Revision B" - both false on direct inspection of the PDF |
| P3 | Attenuator package moves 0603 -> 0805: PTFR0805Q200KN9 (C23067436, RESI), 200 kohm 0.02 pct 10 ppm/C. The 0603 row at this exact grade returns an unfetchable datasheet on live search, and an unverifiable part is not a part. The 0805 row is datasheet-backed with deeper stock (6235 vs 4150). Mechanical change only - flagged for P4 symbol/footprint and P6 placement | Same electrical spec, and the boards whole accuracy argument rests on these five parts, so a row we cannot read the datasheet for is disqualified by the roles own rule. Larger pads are also mildly helpful on a guarded high-impedance node |

| Phase | What | Why |
|---|---|---|
| P3 | U1 ADS8326IBDGKR (C544731, TI, 12.39 USD) is SINGLE SOURCE with NO pin-compatible alternate, recorded as an explicit empty alternates list rather than a fabricated one: the datasheets own pin-compatible family members are mostly 12-bit or typical-spec and do not close the error budget. BOM totals 155.16 USD at qty 5 (31.03/board), 84 pct of it U1 + U2. Tightest stock margin is ADR4520B at 314 pieces (12.6x the floor), hedged by the A grade at 908 | The alternate field exists to record real second sources; filling it with a part that cannot meet spec would convert a known risk into a hidden one. A Tokmas-branded ADR4520A clone row was found and REJECTED - its own attributes (2.8-40 V supply, 20 mA output, 8 ppm/C) do not match the genuine part, which is the brand-pinning rule doing exactly what it was written for |
| P3 | CORRECTION - an unsourced claim of MINE was caught by the extractor and is withdrawn: the ADR4520 datasheet does NOT say solder-heat-shift magnitude depends on board size, thickness or material. Rev 0 (4/12) ties SHR only to package material and solder process (leaded vs lead-free); the extractor grepped all 32 pages for those terms near solder mentions and found nothing, then flagged rather than fabricated. The board-size dependency belongs to a DIFFERENT vendors document (the TI material in the same research task), and I relayed it into the spawn prompt as if it were ADIs | This is exactly the laundering failure the whole research leg exists to prevent, committed at the relay step rather than in a record: two vendors documents were read in one research task, and when I compressed that task into a spawn prompt I attached one vendors caveat to the other vendors part. The records themselves are clean - the error appeared only in my prose. Consequence: the U2-vs-mounting-hole keep-out was ALREADY labelled as our inference and stays, but its stated basis must shrink to what is real - a specified reflow shift of +/-0.02 pct typ, as large as the whole B-grade initial accuracy - with no vendor claim about board flex behind it |
| P3 | U2 land pattern: the ADR4520 datasheet contains NO recommended land-pattern drawing - only package outline dimensions (p32) - so pad_size_mm and row_spacing_mm are deliberately unset in the extraction and the footprint must come from a standard JEDEC SOIC-8 narrow-body pattern. Pitch 1.27 mm is stated and identical either way. No exposed pad. Pin 8 is TP, DO NOT CONNECT; pins 1/3/5/7 are NC | Recorded so the librarian does not go looking for a vendor land pattern that does not exist, and so a schematic agent does not wire the TP pin. The extractor correctly refused to derive pads from the package outline, which is the rule that stops a footprint being built from the wrong drawing |

| Phase | What | Why |
|---|---|---|
| P3 | NO series/current-limiting resistor between the reference output and the converters REF bypass, despite the ADS8326 datasheet offering one "in front of the capacitors". The condition it attaches does not bite here, and fitting it would COST accuracy rather than buy it | Four reasons. (1) The guidance is conditional on the reference not being a dedicated low-impedance source; the ADR4520s stable load-capacitance window is 1-100 uF two-ended, so the 47 uF REF bypass is inside spec and the STABILITY motive for the resistor is absent. (2) It would actively hurt: the SARs reference current VARIES WITH INPUT CODE (TI SNAA320B), so any series R turns that varying current into a code-dependent voltage error - a NONLINEARITY, the worst class of error on a DC-accuracy board - where at 7 uA and 4.7 ohm it is ˜33 uV = 16 ppm = 0.08 mV at the terminal. Without the resistor the reference own output impedance makes the same term nanovolts. (3) TIs own application circuit for this converter (fig 42) drives the 47 uF bare from an amplifier with no series element. (4) Power-up inrush is bounded by the references own +/-10 mA output rating - ˜10 ms to charge 47 uF - which the host >=10 ms wait already covers. C5 stays as the references own local output cap at its pin and C3 as the SARs reservoir at the REF pin; in parallel they total ˜48 uF, still inside the window |
| P3 | CONFIRMED, not a new risk: the error budget already used the CORRECT converter specs. At a 3.135-3.465 V rail the board sits in the ADS8326s VDD = 2.7 V characteristics table, not the 5 V table, and the 2.7 V table is looser for the IB grade - INL +/-2.5 LSB (vs +/-1.5) and gain +/-16 LSB (vs +/-12). Those are exactly the numbers P2 budgeted | The extractor flagged the two-table split as potentially relevant to the budget, which was the right thing to raise. Checking it: the architect quoted offset +/-1 mV, gain +/-16 LSB and INL +/-2.5 LSB, i.e. it had already taken the conservative table for our rail. Recording the confirmation so a later reviewer does not reopen it. Related: DCLOCK max is 4.8 MHz in the 2.7-3.6 V class rather than 6 MHz - irrelevant at our 500 kHz |
| P3 | MAJOR RISK, named and bounded, not closed: TIs own "Driving ADS8326" reference circuit (OPA320 datasheet fig 44, p25) - our EXACT amplifier-plus-converter pairing - warns that single-supply operation "loses a small number of ADC codes near ground" and shows a -0.3 V V- generator as the fix. This board ties V- straight to GND and its input span STARTS at 0 V, so the bottom of the range is precisely where the warning bites | The DC arithmetic says we are probably fine and the TRANSIENT question is open. OPA320s swing spec is 10 mV typ / 20 mV max from the rail at RL = 10 kohm, which is ˜164 uA of load; this boards DC load is ˜1 uA (the C_FLT recharge), so the static drop scales to ˜61 uV at the ADC input = ˜153 uV at the terminal, about 3 pct of budget. What does NOT scale is the transient: near ground the output stage has minimal headroom, its output impedance is highest, and it must still absorb the per-conversion charge kick inside t_acq. That is a settling question, not a DC one, and it is exactly what the architect already named the 'zero-scale-swing' bench for (<= 1.0 mV above GND at 0 V input) together with 'acquisition-settling' run AT THE BOTTOM OF THE RANGE |

| Phase | What | Why |
|---|---|---|
| P3 | CONTINGENCY, pre-authorised so it is not re-litigated under time pressure at P8: if the zero-scale bench shows lost codes near ground, the -0.3 V generator is DATASHEET-REQUIRED SUPPORT and not an excluded feature - it appears in the vendors own recommended circuit for this converter-amplifier pair. It would still be a second rail, so it becomes a mode-boundary item that goes to the OWNER as a decision, exactly as requirements.md section 1 anticipated | The scope tier excludes a second rail the block does not need IN ORDER TO WORK. If the block cannot measure 0 V - a stated part of the input span - without it, then it needs it to work, and the exclusion does not apply. Recording the reasoning now, while it is a hypothetical, keeps the ruling honest |
| P3 | OPA320 PASSES the isolation-resistor screening test decisively, and the flat-R_O assumption that was UNVERIFIABLE for the OPA333 is verified for this part: R_O is 90 ohm typ, and TIs text states it "remains very flat over a wide frequency range", confirmed visually on fig 19 as ˜90-100 ohm from low frequency to roughly 1-10 MHz. Settling ceiling 750 ohm at C_FLT = 1 nF versus a stability optimum of only ˜27 ohm at R_O = 90 ohm and GBW 20 MHz - a wide legal interval where the OPA333 had none. TI separately recommends 10-20 ohm isolation for this part; R6 is provisionally 49.9 ohm pending the bench | This is the same screening test recorded at P2, now run against the part actually sourced, and it inverts cleanly: the OPA333 failed it on a single-frequency R_O we could not confirm was flat, while the OPA320 both passes it and publishes the flatness the method assumes. Worth noting the two parts differ by more than 20x in R_O (90 ohm vs 2 kohm), which is most of why one is a SAR driver and the other is not |
| P3 | GAP carried into P8, not bridged: OPA320s settling is characterised only at VS = 5 V with CL = 50 pF, and its tightest PUBLISHED figure is 0.0015 pct = 15 ppm and simulation-derived - about 2x looser than this boards ˜7.6 ppm target, at a different supply and a 20x smaller load capacitance. No datasheet number covers our operating point | The extractor flagged it rather than extrapolating, which is right. It means the acquisition-settling bench is not a confirmation exercise - it is the only evidence that will exist for this parameter, and it must be run at 3.3 V with the real C_FLT and at the bottom of the input range where the zero-scale risk also lives. Two open questions, one bench |

| Phase | What | Why |
|---|---|---|
| P3 | J1 DRILL MUST BE ENLARGED 1.30 -> 1.50 mm per the vendors own recommended PCB hole. Orchestrator APPROVES this footprint hand-edit (the role correctly refused to make it unilaterally). The pad must grow with it to keep at least the JLC 0.15 mm annulus, so pad diameter >= 1.80 mm | This is not a margin question, it is a fit question. WJ500V-5.08-class screw terminals carry ~1.0 mm SQUARE pins, whose DIAGONAL is 1.41 mm - larger than the pulled 1.30 mm round hole. A part that does not physically enter its holes is a scrapped board, and it would not be discovered until assembly. The librarian fetched the Kangnex mechanical drawing and read 1.50 mm directly, so we are following the vendors number rather than estimating. Note the failure mode the annulus check alone would have missed: 1.30 mm CLEARS the JLC annulus floor comfortably, so every automated check passes on a hole the pin cannot fit through |
| P3 | U1 VSSOP-8 land pattern ACCEPTED as pulled; the reported 42 pct row-spacing mismatch is almost certainly a MISREAD of the vendor drawing, not a footprint defect. Arithmetic: DGK/MSOP-8 lead span is 4.90 mm, so lead centres sit ~2.17 mm from package centre. The pulled pads are at 2.13 mm - a 0.04 mm match. The "implied 3.0 mm centre-to-centre" would put pads 1.5 mm from centre, i.e. UNDER the package body where no lead reaches | The 3.0 mm figure appears to come from subtracting a 1.4 mm pad length from a 4.4 mm OUTER-EDGE span, which mixes two different measurement conventions. fp_verify passed on count, pitch and pin 1 with only a pad_size warning, the footprint is LCSCs own for the exact LCSC part being ordered, and JLCs assembly process is matched to it. Accepting - but the discrepancy is recorded so P9 DFM looks at it rather than inheriting it silently |
| P3 | ACCEPTED with follow-ups: (a) the ADR4520 symbol hand-edit setting pins 1/3/5/7/8 to no_connect STANDS and must be RE-APPLIED after any re-pull or re-run of lib_pin_types.py, which has no NC concept and was dry-run-confirmed to silently revert all five to passive; (b) U3s missing F.SilkS pin-1 dot goes to P6 silk, where the tooling for it lives - pad asymmetry conveys orientation meanwhile; (c) U3 row spacing 2.60 vs 2.70 mm and U1s courtyard not enclosing its own pads are both accepted (the latter is the known systemic easyeda2kicad pattern already compensated downstream); (d) J2 geometry accepted on internal consistency alone - no manufacturer datasheet exists | Item (a) is the one with teeth: it is a silent-revert trap, so it is recorded as a standing re-apply obligation rather than a completed action, and flagged as a lib_pin_types.py enhancement candidate. The rest are cosmetic or already-known and none blocks P4 |

| Phase | What | Why |
|---|---|---|
| P3 | coverage-mapper NOT spawned at P3 exit despite a mapping_request, and no further research opened. Both remaining research tasks stay unspent. P3 coverage: all THREE part slots COVERED, B2/B3/B4 provisional, B1 gap on constraints-emission alone - the gap already recorded at P2 as designed-under | Checked rather than assumed. Every provisional class names 'maturity-below-floor' as its blocker with the CORRECT records already matched to the right slot and class (e.g. B2 feedback -> precision-buffer-capacitive-load-isolation and -riso-settling-stability-collision). So these slots are limited by MATURITY, not by missing record-to-slot edges, and a mapping pass cannot lift them - only owner approval can, and promotion is deliberately deferred until the sibling runs finish. B1s one unmet class has no candidate record BY CONSTRUCTION: the attenuator researcher declared constraints-emission an honest gap and wrote nothing for it, so there is no edge for a mapper to find. Spawning one would produce a file that changes no verdict |
| P3 | P3 part-level coverage needed NO research: all three IC extractions produced substantive layout_notes and each part slot reads COVERED. The seeding flags (–maturity-floor proven –research-provisional) were run first per the learning-target contract, and they re-escalated the four already-researched BLOCK slots back to gap; that is the flags working on maturity, not a new finding, so the authoritative P3 report is the re-run without them | Worth recording because the flagged run LOOKS alarming - four blocks back at gap after a full research leg - and a less careful reading would have opened four redundant tasks against a budget of two. The contract itself says to re-run without the flags after research; the subtlety is that at P3 the FIRST call of the phase still carries them, so the same re-escalation happens a phase later on slots that were never in question |
| P4 | GUARD RING NET, stated explicitly because getting it wrong is silent: the driven guard around the divider tap and U3 +IN belongs on the BUFFER OUTPUT net, NOT on /AIN_DIV. A ring on the same net as the node it surrounds is not a guard - it IS the node - and provides no guarding at all. U3 is a unity-gain follower, so its output sits at the tap potential with low impedance, which is exactly what a driven guard needs. No schematic element is required; it is a P7 copper instruction on the buffer output net | constraints.json records the guard as an area reservation along the /AIN_DIV corridor, which correctly says WHERE the ring goes but is silent on WHICH NET it carries. A layout agent reading only that could reasonably pour the ring on /AIN_DIV and produce something that looks like a guard ring, passes every check, and does nothing. Magnitude for scale: at the 240 kohm Thevenin, 1 nA of surface leakage is 240 uV at the tap = 600 uV at the terminal, 12 pct of budget |

| Phase | What | Why |
|---|---|---|
| P4 | CLARIFICATION of the -IN sense requirement, correcting my own earlier framing: -IN and the attenuator string bottom are ONE NET (GND) and MUST be - the divider bottom IS ground, and -IN carries a -0.3/+0.5 V common-mode bound relative to DEVICE ground, so a separately-floated analog ground would risk violating it. The Kelvin intent is therefore a PLACEMENT AND GEOMETRY requirement, not a netlist one, and the schematic as built is correct | I had said "wire it as its own net", which was wrong and would have produced a worse board. What actually has to hold at P6/P7: R5s bottom pad sits IMMEDIATELY ADJACENT to U1 pin 3, the copper between them is short and direct, and NO digital or supply return current flows through that copper. The error term is the IR drop between those two points, and with a solid pour plus adjacent placement it is microvolts; with careless placement, SPI return current crossing that segment puts millivolts there, and 1 mV becomes 2.5 mV input-referred. constraints.json already carries this correctly as an R5->U1 corridor on net GND with the "reference tie, not a ground connection" rider. Note what this means for routing: with a B.Cu GND pour the router will connect both pins to the pour rather than draw a dedicated trace, and that is FINE - the requirement is on the geometry, not on a trace |
| P4 | E1 ACCEPTED and my own P4 clarification is REVERSED: placement alone does NOT preserve the Kelvin sense. On the netlist U1.3 is one of 16 GND nodes, so the B.Cu pour will thermal-connect pin 3 exactly like pin 4 and reference the measurement to the pour AT THE CONVERTER - deleting the 2.5x cancellation with every gate green. A corridor is a keep-clear swath, not a connectivity rule. FIX: give the sense its own net /AGND_SENSE (U1.3 + R5 bottom) tied to GND through a single 0 ohm link R8 placed at the string bottom | I had reasoned that adjacent placement makes the IR drop microvolts and called the netlist-level fix unnecessary. That is true of the drop but misses the mechanism: the pour does not merely add a parallel path, it TERMINATES pin 3 locally, so the sense node stops being the string bottom at all. The 0 ohm link makes the arrangement netlist-enforced - every gate then checks it, the pour cannot cover /AGND_SENSE, and the router must route U1.3 to R5s pad. Cost is one Basic 0402. Algebra after the fix: the measurement is K*(V_sig - V_S) + Vos, so the ONLY residual is the offset between J1s return pin and the string bottom, which the reviewer measured as sub-1 uV on a solid pour. In scope: this is a topology element the block needs to work, not protection, filtering, an indicator or a second rail |
| P4 | W1 (reference headroom vs ripple) ACCEPTED AS ANALYSIS, no design change. The 87 mV DC margin stands and ripple troughs are NOT a dropout risk here: the reference carries ˜49 uF on its output against a microamp load, so a trough of tens of mV lasting microseconds cannot move VOUT measurably. No RC filter is added to the references input - the scope tier excludes filtering the datasheet does not require, and this analysis says it is not required | The reviewer is right that the 87 mV was compared against DC tolerance alone while the verified record says tolerance + IR drop + ripple trough, and that is the correct challenge to raise. But the output capacitance answers it: a series-reference dropout spec is a DC condition, and the loop plus 49 uF holds VOUT through brief input dips. What DOES bind, and is now stated explicitly: this design is validated for a host rail whose DC level stays at or above 3.135 V. If the real rail sits lower, the margin is gone and no capacitor recovers it |

| Phase | What | Why |
|---|---|---|
| P4 | W2/W3/W4/W5 dispositions: W2 - the published error budget was computed on the OPA333 and must be restated for the OPA320 actually fitted (RSS 3.25/3.42 mV, worst case 8.58/10.74 mV; over-temp worst-case margin 14 pct -> 10.5 pct, and the bias-current row improves ˜200x, never credited). W3 - the host power-up wait rises from >=10 ms to >=50 ms: slew-limited charging of 49.2 uF at the 10 mA source limit is ALREADY 7-10 ms, so the old figure had ˜1x margin, not 50x. W4 - J2s 180-degree mis-mate shorts the host rail to its own ground; the connector checklist requires this be STATED when destructive, so it goes on the silkscreen and into the design doc. W5 - C7s 20x-C_SH floor was checked on NOMINAL; the fitted J-grade part guarantees only 950 pF against a 960 pF floor, so the P8 bench must choose C7 on its GUARANTEED minimum, with R6 present (the isolation resistor is what lifts the pure-cap-load limit) | None of these changes the topology; all four are numbers or statements that were wrong or missing. W3 is the one with real consequence - a host that waits 10 ms would read a reference that has not finished starting, and every conversion in that window is simply wrong. Also noted for correction: constraints.json records VREF current as 7 uA where the datasheet row is 14 uA max; nothing moves, flagged so it is not reused somewhere tighter |
| P8 | ZERO-SCALE RISK CLOSED BY MEASUREMENT - the -0.3 V generator pre-authorised in the P3 contingency is NOT needed and will not be fitted. DC swing at 0 V input is 4.50 uV, exactly R_O x I_leak, = 0.090 ADC CODES against the vendors warning of "a small number of codes". Transiently the sampling cap arriving at VREF pushes the ADC node to +83.9 mV and the buffer output to +54.0 mV, and the SAMPLED value is back within 10 nV of ground, recovered in 1.24 us of the 9.0 us window | This was the largest open risk on the board and the only one that could have forced a second rail and an owner mode-boundary decision. It is now settled by a bench that models the real switched-capacitor event rather than a resistor, with a seeded defect (V- lifted to +0.2 V) confirmed to trip the bound from 4.5 uV to 200 mV. The DC arithmetic recorded at P3 predicted ˜61 uV by load-scaling; the measurement is 4.5 uV, so the estimate was conservative by an order of magnitude and the conclusion holds by a wide margin |

| Phase | What | Why |
|---|---|---|
| P8 | W5 CLOSED, C7 STAYS AT 1 nF. At its GUARANTEED 950 pF the node settles to 0.136 uV in 0.149 us - 115x inside the 15.6 uV budget and 60x inside the 9.0 us window, and indistinguishable from 1000 pF nominal. The 1 pct shortfall against the 20x-C_SH floor has no settling consequence. The 2.2 nF C0G candidate also clears (0.143 uV, 0.643 us, PM 75.2 deg) but buys only a halved droop for 4.3x the settling time | The 20x rule is a sizing heuristic for how much of the sampling kick the reservoir absorbs, not a settling threshold, so a 1 pct miss on it means nothing once the actual settling is measured. Choosing the larger cap on heuristic grounds would have traded a real 4.3x in settling time for a nominal improvement in a term already 115x inside budget |
| P8 | ACCEPTED AS A DOCUMENTED BEHAVIOUR, not a defect: a FULL-SCALE STEP at the terminal takes 25.5 us to settle to 15.6 uV, against the 9.0 us the architects bench spec named. The mechanism is intrinsic - the 240 kohm tap Thevenin charging the buffers own input capacitance - and it is 11.3 us even at ZERO board stray, on the amplifiers 4 pF Ccm alone. Consequence: the first conversion after a full-scale terminal step is invalid and the next one is fine | The 9.0 us figure was the wrong yardstick. t_acq bounds the per-conversion charge event on a MULTIPLEXED front end; this is a single-channel DC voltmeter with no mux, so the time actually available after an input step is the 100 us sample period, and 25.5 us fits inside it four times over. Reaching 9.0 us would require a lower Rtot, which the source-loading budget forbids - the same coupled decision the whole board is built around. Bounded at 100 us as a warning and written into the design doc as a host behaviour: discard the first sample after a large input step |
| P6 | Hand-placed canonical layout KEPT; every anneal candidate REJECTED. The free runs best (0.7 pct hpwl gain) put C3 3.97 mm from U1s REF pin against a 2.5 mm limit, C2 2.44 mm against 2.0, stretched the /AIN_DIV stub from 2.60 to 7.18 mm, moved C7 from 1.97 to 12.01 mm off the ADC input and scattered the mounting holes. The locked run left movable_clusters 0 and made hpwl worse (242 -> 260) | Root cause found and recorded in the repo learnings: place_anneal RE-DERIVES satellite slots from place_seed rather than inheriting the placement it was handed (hpwl_input 242.12 vs hpwl_start 299.80). So on a board whose value is in a few short critical stubs - a 240 kohm tap, a 47 uF at a REF pin, a sense tie - the annealer optimises a global metric by destroying exactly the local geometry that matters, and a 0.7 pct hpwl win costs a 60 pct violation of a declared decoupling distance. Final: hpwl 327.23 -> 241.87, route probe completion 1.00 (41/41 nets), place gate 0/0 |
| P6 | All 25 footprints LOCKED deliberately after placement. Accepted with its known cost: a future place_metrics run gates keepout and non-declared-edge containment on is_movable, so those legs go quiet on a fully locked board. Verified by hand instead - 0 rule areas present, and the union of all extents is within the outline | The guard-ring room, the tap stub, the sense tie and the string order are SPECIFICATIONS that no gate can re-verify: nothing in the check suite knows that /AIN_DIV must stay short or that R8 must sit at the string bottom on the front side. Locking is the only mechanism that stops a later automated pass from optimising them away, which is precisely what the anneal candidates tried to do |

| Phase | What | Why |
|---|---|---|
| P6 | Outline FIT applied: 63.95 x 66.03 -> 64.9 x 34.92 mm, area 4223 -> 2266 mm2. Place gate re-passes 0/0. Height is fully earned; the WIDTH is being re-earned in a scoped second pass because ˜20 mm of it is the J1->R1 corridor at 16.5 mm, and that number was set by EDGE_TOL against the PROVISIONAL outline rather than chosen | The separation itself is defensible and stays - a screw terminal carrying field wiring should not sit beside a 1 Mohm node, and letting the ROBUST raw-input net carry the span instead of a sensitive one is the right structural choice. What is not defensible is the specific 16.5 mm, which nothing argued for. Under a canonical binding the size must be a RESULT WITH A REASON; accepting an artifact of the room the placement was handed is the same failure as bb-buck, just in the other direction |
| P7 | Two P7 routing constraints handed forward by placement: (1) the SPI fan needs exactly 2 crossings (/CS x /SCLK and /CS x /DOUT) because J2s pin order is CS,SCLK,DOUT while the converters is CS,DOUT,SCLK - keep the via pairs EAST of U1, out of the analog zone; the J2 rotation that removes them costs 3 rail crossings instead, and a schematic-level J2 pin swap would remove them entirely at a later revision. (2) The guard ring cannot close through the U3 pin2/pin3 channel - it is 0.35 mm and cannot carry 0.127 mm of trace plus 2 x 0.127 clearance - so the closed ring must encircle pin 2 (V-) TOGETHER WITH pin 3, with pin 2 escaping straight down through its own via to the pour | Both are geometry facts discovered while placing, and both would otherwise be rediscovered by the router as failures. (2) matters most: a guard ring that cannot close is not a guard ring, and the fix - enclosing the V- pin inside the ring - is counter-intuitive enough that a router would not invent it |
| P6 | BOARD SIZE EARNED: 54.75 x 34.92 mm, 1912 mm2. The J1-to-R1 separation is 7.3 mm and decomposes into three things that each have a reason and nothing else: 4.5 mm is the screw terminals OWN BODY (courtyard east of its pad row - that span is connector, not clearance), 2.0 mm is clear board for screwdriver access with field wire already dressed, and 0.8 mm is R1s pad approach. Only the 2.0 mm is discretionary | The earlier 16.5 mm was not merely unearned, it was WORSE on the mechanism the knowledge records actually name: leakage and pickup both scale with the input traces length, so every record touching this net argues for shorter. EDGE_TOL chose 16.5; a connector body and a screwdriver choose 7.3. Also found while moving: H1/H2 - not J1 - were defining the west edge at x 9.05, so moving J1 alone would have earned nothing, and H3 was 0.2 mm proud on the east so J2 rather than a mounting hole now defines that edge. The node that genuinely wants distance from a torqued terminal is the 240 kohm tap (called microphonic in the architects group note), and it sits 12 mm further east, outside the flex zone. No thermal or creepage argument was claimed, because at 36 mW and 5 V neither exists |

| Phase | What | Why |
|---|---|---|
| P7 | CORRECTION of my own routing brief: +3V3 is widened to 0.5 mm per constraints.json, not left at the Default 0.2 mm I instructed. I told the router "do not force widths" on an IPC-2152 CURRENT argument; the constraints _why anticipates and rejects exactly that argument - "IPC-2152 width at 11 mA is a hairline, so heating is NOT what binds this net - impedance to the decoupling caps is. Route >= 0.5 mm." Different physics, and the constraints is the applicable one | The router followed my brief over a written constraint and FLAGGED the conflict rather than silently resolving it either way, which is the correct behaviour for a subagent. Cost of the error had it stood: ~0.6 mV of extra DC drop, which the error budget does not notice because the conversion is referenced to VREF - but it would also have left a live contradiction between the board and its own constraint file for P8/P9 to trip over |
| P7 | KEPT as a cited deviation: the VREF segment C3.1->U1.1 is routed at 0.6 mm rather than the 0.2 mm Default, citing the verified record sar-adc-reference-charge-loop (wide copper out and back on the reference charge loop). It was PRE-ROUTED by hand specifically so Freerouting could not place a via between the reference bypass cap and the REF pin | A via in that path inserts inductance into the exact loop the record is about - the per-conversion charge that C3 must supply and the reference must replace. Pre-routing a short critical path to deny the autorouter the chance to spoil it is the same discipline as locking the placement, and for the same reason: the tool optimises a metric that does not know what this copper is for |
| P8 | P8 ITEM logged so it cannot fall through: the driven guard ring needs SOLDER-MASK RELIEF over its copper (constraints + the verified record resistive-attenuator-high-z-tap-guard-and-leakage). Routing cannot emit mask openings, so P8 owns it | The point of exposing the guard is that any surface contamination or moisture path leaving the high-impedance tap meets bare guard copper AT THE SAME POTENTIAL, rather than bridging to something at a different one. A guard ring buried under mask still guards electrostatically but forfeits the surface-leakage half of its job, which on a 240 kohm node is the half that was worth having |
| P8 | GUARD-RING SOLDER-MASK RELIEF DEFERRED, not implemented, with the arithmetic for why it is acceptable at THIS impedance. No pipeline script can emit a mask opening (board_edit does outlines, planes_gen does copper, and nothing else touches F.Mask), and hand-editing the board file is forbidden. Recorded as a documented deviation in the fab notes rather than silently dropped | Quantified rather than waved away. For guard-defeating leakage to matter here it must produce more than ~0.2 mV at the 240 kohm tap (0.5 mV at the terminal, 10 pct of budget), which is 0.83 nA, which from a 3.3 V aggressor needs a leakage path BELOW ~4 Gohm. Intact solder mask on a washed board is orders above that, and the ring already intercepts electrostatically regardless of mask. Mask relief is the third layer of defence on a node two orders less sensitive than the 10 Mohm-1 Gohm regime the source record was written for. It would be REQUIRED, and this reasoning would not hold, on a megohm-class divider - which is exactly the design this board rejected on other grounds |

| Phase | What | Why |
|---|---|---|
| P7 | RETURN-PATH FIX by EXCISION, not by shortening: the /CS crossing was rebuilt from a 4.39 mm 45-degree single-via tunnel into a 1.93 mm PERPENDICULAR VIA PAIR at x=66.0, placing BOTH crossings inside the transition vias 0.95 mm excision disks (via_radius + 0.65). /SCLK cleared entirely (was 0.73 mm2 / 0.64 mm error), /DOUT warning cleared entirely, /CS reduced 91 pct in area and 98 pct in crossing length | My proposed fix - move the crossing under bare board - is TOPOLOGICALLY UNREACHABLE and the router showed why: J2 orders CS,SCLK,DOUT while the converter orders CS,DOUT,SCLK, a 3-cycle, so /CS must cross BOTH other signals, and a crossing can only exist where /CS is on B.Cu. The real lever was found by reading why /DOUT was already passing: severity keys on centerline crossing length with a 0.01 mm threshold, so SHORTENING never clears anything - what clears it is excision. And it is physics, not checker-gaming: where the void under the aggressor is a vias own antipad disk rather than a slot, return current detours around a small disk instead of running the length of a cut. It also halved the pour slot |
| P7 | REFUSED: driving the /CS residual to zero by shortening the tunnel 0.09 mm so the two excision disks overlap. It would change NO physics and would cut via-to-trace DRC margin to ˜0.04 mm. Waived instead, with both reasons recorded | This is the clearest example on the board of the difference between fixing a defect and satisfying a checker. The 0.03 mm residual is the gap between two bookkeeping disks; closing it buys nothing electrical and spends real manufacturing margin. The router declined to do it unprompted and asked - which is the right instinct, and the answer is no |
| P7 | PIPELINE DEFECT affecting EVERY ai-ee board, and my own diagnosis of it was wrong: –verify-fill does NOT use an independent fill model. assert_fresh runs real kicad-cli –refill-zones on a temp copy, and geom._refill_copy stages the .kicad_pro but NOT the .kicad_dru. Measured three ways: pcb+pro+dru gives guard 17.176 mm2 and GND 1745.233 mm2, EXACTLY the committed fill; pcb+pro gives 6.613; pcb alone gives 6.613. The .kicad_pro changes nothing - the .kicad_dru is everything, because without it the 0.127 mm clearance floor is gone and KiCad refills at its stock 0.2 mm | rules_gen writes a DRU for every board in this pipeline, so every board is affected, not just ones carrying a connect:solid zone as I had guessed. The READ path is safe (it only produces a spurious StaleFillError), but the same omission in a WRITE path would commit a silently wrong fill. The router verified planes_gen, route_auto and route_edit all stage the DRU and are unaffected. I diagnosed this as a checker-vs-KiCad fill-model disagreement; it is a missing file in a temp-copy helper |
| P7 | DESIGN MARGIN DISCOVERED, and it is a real one: the guard ring CLOSES at the 0.127 mm clearance floor and DOES NOT CLOSE at 0.2 mm. At 0.2 mm the pour drops 19.03 -> 8.68 mm2, the entire south leg is culled and the closure test finds ZERO holes - yet it still fills, and still looks like copper in a render. Any change to the clearance floor, the fab class or the copper weight MUST re-run the closure proof | Found accidentally while measuring the DRU defect above, and it is the most fragile thing on the board. The guard ring is the only defence for a 240 kohm node, its function depends on ENCIRCLING that node, and its encirclement survives on a 73 um margin against the fab floor. Worse, every cheap check passes when it fails: DRC is clean, the zone fills, the render shows a ring. Only a containment test on the filled polygons can tell the difference, which is exactly why the router was asked to prove closure geometrically rather than visually |

| Phase | What | Why |
|---|---|---|
| P8 | E1 CONFIRMED REAL against the vendor drawing: J1s wire throats face EAST, into the board. The Kangnex WJ500V-5.08 drawing shows the body is ASYMMETRIC about the pin row - 4.50 mm on one side, 3.14 mm on the other - and the wire-entry face is the 4.50 side, which is exactly the geometry the reviewer used for its transform. J1 must be rotated 180 degrees and /AIN_RAW + GND re-routed | Two agents disagreed and the tiebreak matters: placement proved the direction from an orthographic render plus the footprints silk entry arrows; the reviewer proved it from the 3D models actual opening geometry (two 3.0 mm openings, z 3.0-9.8 mm, on the y=+4.50 face, the other face solid) plus the rot-90 transform. The reviewer ALSO independently found that the footprints silk arrows sit on the WRONG FACE - which is precisely what misled the render check. So the two sources inside the footprint contradict each other, the vendor drawing agrees with the 3D model, and a render of a footprint cannot be more authoritative than the part geometry it draws. Consequence if shipped: the field lead must be inserted from the boards interior and laid back over the 1 Mohm string and the J2 ribbon - unusable as intended |
| P8 | E5 ACCEPTED as a real defect introduced by the guard-ring closure workaround: GND conductors sit INSIDE the closed ring. U3 pin 2 and a GND via at (41.850, 39.850) are 0.227 / 0.289 / 0.350 mm from the guarded node with the guard OUTSIDE all three, so the ring cannot intercept those leakage paths. Priced on blocks.md own 1 Gohm model that is 1.20 mV at the terminal against the 0.78 mV the guard was built to stop, and budget row 9 carries 0.10 mV - understated ˜12x. The via is MOVABLE and will be moved out; the 0.350 mm pin2/pin3 channel is irreducible at the 0.127 mm floor | This is the cost of the closure workaround we chose deliberately at P6 - encircling V- together with the input pin - and nobody priced it until now. The spec still closes (RSS 3.23 -> 3.44 of 5.00 mV), so it is not a redesign, but it is a real error term that was invisible in the budget. Moving the via is free and removes the worst of the three paths |

| Phase | What | Why |
|---|---|---|
| P8 | E5 FIXED BY RE-ROUTING THE GUARDED NET, not by moving the offending via - the via could not be moved out because the pad it serves is inside the ring, so anything touching it is inside. /AIN_DIV is a 3-pad net whose spanning tree can hang the stub off EITHER resistor for the same 2.600 mm; moving it from R3.2->U3.3 to R4.1->U3.3 put the stub on the east side of the pocket and deleted both sub-0.35 mm paths at zero length cost. GND-inside-ring distances go 0.227/0.289/0.350 mm to 0.350 (pin 2, irreducible) and 0.556 (via); scaled on the same 1/d model that is ˜1.20 mV -> ˜0.52 mV, under the 0.78 mV the guard exists to stop | Worth recording as a pattern, not just a fix: when a keepout cannot be cleared by moving the intruder, check whether the PROTECTED object has freedom instead. A net with more than two pads has spanning-tree freedom that costs nothing in length, and here it bought the whole improvement. The instruction I gave - move the via - was unreachable; the agent found the reachable version |
| P8 | E1 fix verified from the FOOTPRINTS OWN GEOMETRY rather than from my ruling: pads sit on local y=0, F.Fab spans local y -5.64..+4.52 and F.CrtYd -5.5..+4.5, so the 4.50 mm face is local +y, and KiCad maps local +y to board +x at rot 90 - throats east, exactly as the reviewer found. After rotating to rot -90 the courtyard reads x[19.955, 30.045] about a pad row at x=24.500: west face 4.545 mm, east 5.545 mm. Also learned the hard way: A 180-DEGREE ROTATION SWAPS THE PADS, so /AIN_RAW moved from y 41.600 to 36.520 and GND took its place; both were re-routed fresh | The agent re-derived the direction from an asymmetry number rather than accepting my instruction or looking at another picture, which is the correct response to a defect whose root cause was trusting a picture. The pad swap is the kind of consequence that turns a ”just rotate it” instruction into a wrong board if nobody re-routes - and place_edit REFUSED the move on a routed board and demanded a rip first, which is the guard working |
| P8 | Board size held at 54.750 x 34.920 mm after the J1 rotation. A re-fit was checked with –report-only and would have given 54.75 x 34.937 - a 17 um delta, below manufacturing noise - and was REFUSED because applying it would trigger the board-grew path and force a planes_gen re-run for nothing | Re-fitting is not free: it invalidates the pour and drags a re-verification of the guard ring, which on this board closes on a 73 um margin. Spending that to chase 17 um is a bad trade, and checking with –report-only before touching anything is exactly the right way to find that out |

| Phase | What | Why |
|---|---|---|
| P8 | W1 root cause fixed rather than its symptom: verify_all warned "no bulk on +3V3" and it was a FALSE POSITIVE - C1 is a 10 uF sitting 0.925 mm from J2.1 - but the cause was that decoupling.json OMITTED C1 entirely, so check_decoupling never examined the rail-entry bulk at all. C1 is now associated, and both check_decoupling and check_pdn read 0 | The warning was right to fire and wrong about why. A check that passes because an association is missing is worse than one that fails: it reports on copper it never looked at. Fixing the sidecar makes the check examine real copper instead of passing by omission |
| P8 | ACCEPTED, with the deviation disclosed: the J1 silk arrows were corrected on the board instance by a one-off SWIG worker, because NO PIPELINE OP OWNS FOOTPRINT-INSTANCE GRAPHICS. add_text/remove_text/move_text are the only silk ops and remove_text matches board.GetDrawings() - board-frame gr_text - so it cannot see footprint children at all; route_edit does tracks and vias, board_edit does Edge.Cuts, and fpfix takes –lib <dir>.pretty and never a board. The worker followed place_edits own contract: stage a copy beside the board, edit under bundled python, re-parse and DRC the copy, os.replace only after it verified clean | The alternative was shipping a board whose own silkscreen contradicts its own labels within millimetres - an arrow pointing at a solid plastic wall next to text saying SIG. The agent disclosed the deviation plainly rather than quietly raw-editing, and used the safest available pattern. Recorded as a PIPELINE GAP worth a plan step: a mirror_fp_graphic / remove_fp_graphic op on place_swig. Nice detail found on the way - the correct mirror axis is the BODY axis at x 25.06, not the pad row at 24.5; mirroring about the pad row would have overhung the body by ~1 mm and collided with the new SIG label |
| P8 | Issue 1 (corridor_void cluster, 2 errors) CLOSED as fixed: /SCLK and /DOUT cleared entirely by rebuilding the /CS crossing as a perpendicular via pair so both crossings fall inside the vias 0.95 mm excision disks; the /CS residual fell 2.35 -> 0.20 mm2 and 1.81 -> 0.03 mm and is waived on two independent grounds | Recorded as a decision because the issue record itself takes no note field. One of the two findings was fixed outright, one was reduced by ~91 pct and waived - so "fixed" is the honest status for the cluster while the waiver carries the remainder |

## Sheet plan

## sheets.md - bb-adc sheet plan, refdes ranges, CANONICAL net names

### 1. Sheet plan: ONE FLAT ROOT SHEET

| Sheet | File | Blocks it carries | Interface nets at its boundary | Refdes ranges | ‘pwr_base‘ |
|---|---|---|---|---|---|
| root (only) | ‘kicad/bb-adc.kicad_sch‘ | B1 attenuator, B2 buffer, B3 converter, B4 reference, J1 input, J2 host header, H1-H4 mounting | none - the board’s only interfaces are the two physical connectors (‘/AIN_RAW‘ + ‘GND‘ at J1; ‘+3V3‘, ‘GND‘, ‘/CS‘, ‘/SCLK‘, ‘/DOUT‘ at J2) | ‘U1-U9‘, ‘R1-R19‘, ‘C1-C19‘, |

‘J1-J9‘, ‘H1-H9‘ | **1** (‘#PWR01..‘, ‘#FLG01..‘) |

**A hierarchical split is deliberately REJECTED, and this is a considered deviation from the architect role's default instruction.** The reason is recorded, machine-verified and has already cost two boards on this project: a wiring label inside a CHILD sheet exports to the netlist as /<sheet>/<LABEL>, not /<LABEL> (LEARNINGS 2026-07-22, kicad-sch-api 0.5.6 hierarchy semantics: child-internal nets become /<sheet>/NAME; only nets merged to the root through a sheet pin keep the root-side name). Every constraints.json entry that spells the net /<LABEL> then silently no-ops - check_current finds no copper on a net that does not exist and reports nothing. That is the P4 amendment class that cost lumina-par, and the reason bb-buck and sbuck-5v3a are also flat.

This board has 20 electrical parts and 14 nets (s2's table is the count; the earlier "12" was already one short of its own table). It fits one A4 page with room to draw the guard ring symbolically. The cost of hierarchy here is real and its benefit is zero.

Refdes ranges are therefore trivially unique. They are stated anyway because the rule is "unique ACROSS sheets": **if a later amendment ever splits this sheet, the natural cut is afe (J1, R1-R6, U3, C6, C7) and digital (U1, U2, R7, J2, C1-C5, C8), which take pwr_base = 40 and 80 and keep the prefix ranges above** - and every net name in s2 must then be re-checked against the new export.

## 2. CANONICAL NET NAMES - these are now binding

research/power.json proposed +3V3 / VREF / GND and left the signal names open. P2 fixes all of them here. These are the names the schematic MUST produce and the names constraints.json uses.

| Canonical net | KiCad mechanism | Spans | Declared in constraints.json as |
|---|---|---|---|
| **‘+3V3‘** | power symbol (VALUE = net name) -> exports BARE | J2.1 -> C1 -> **R7** , U2 IN, U3 V+ | ‘power‘ 11 mA |
| **‘VDD_ADC‘** | power symbol with **Value = ‘VDD_ADC‘** -> exports BARE | **R7 -> C2 -> C8 -> U1 VDD.** The converter’s supply pin ONLY | ‘power‘ 3 mA |
| **‘GND‘** | power symbol -> exports BARE | everything; B.Cu pour; J2.2 and J2.6 | ‘power‘ 11 mA ‘plane_fed‘ ‘pdn:false‘, ‘planes‘ B.Cu |
| **‘VREF‘** | power symbol with **Value = ‘VREF‘** -> exports BARE | U2 OUT -> C5 -> C3 -> U1 VREF pin | ‘power‘ 1 mA |
| **‘/AIN_RAW‘** | root-sheet LOCAL LABEL -> ONE leading slash | J1.1 -> R1 | ‘voltages‘ 5 V |
| **‘/ATT_A‘** | root local label | R1 -> R2 | not declared (see s4) |
| **‘/ATT_B‘** | root local label | R2 -> R3 | not declared |
| **‘/AIN_DIV‘** | root local label | R3 -> R4 -> U3 ‘+IN‘. **THE high-Z tap, 240 kohm Thevenin** | not declared - it is a placement/routing constraint, not a width rule (s4) |
| **‘/ATT_C‘** | root local label | R4 -> R5 | not declared |
| **‘/AIN_BUF‘** | root local label | U3 OUT -> U3 ‘-IN‘ (follower) -> R6 -> **and the GUARD RING copper** | not declared |
| **‘/AIN_ADC‘** | root local label | R6 -> C7 -> U1 ‘+IN‘ | not declared |
| **‘/CS‘** | root local label | J2.3 -> U1 CS (active low; the bar is documentation, not part of the net name) | ‘high_speed‘ ref GND |
| **‘/SCLK‘** | root local label | J2.4 -> U1 DCLOCK | ‘high_speed‘ ref GND |
| **‘/DOUT‘** | root local label | U1 DOUT -> J2.5 | ‘high_speed‘ ref GND |

Why VREF and not +2V048: it is a rail feeding a power pin and decoupled at both ends, so it is a power symbol, and a power symbol's exported name is bare. VBUS sets the precedent for a bare non-numeric power net. VDD_ADC follows the same rule for the same reason - a rail feeding a power pin, bypassed at that pin - and it exists because **R7 splits the converter's supply off +3V3, **and KiCad cannot put one net on both pins of a resistor.** A power symbol WINS over a coincident local

label** (LEARNINGS 2026-07-28), so P4 must not mix the two mechanisms on one node, and `component.value = x` is the only way to set a power symbol's Value (`set_property("Value", ...)` is a silent no-op on kicad-sch-api 0.5.6) - which now applies to TWO symbols, `VREF` and `VDD_ADC`.

**Enforcement, not hope:** at P4 run `netlist_audit.py --net kicad/bb-adc.net --constraints architecture/constraints.json`. A net spelled differently in the schematic surfaces as `missing_net` (error).

### 3. Contents of the root sheet

| Refdes | Part class | Net attachments |
|---|---|---|
| ‘U1‘ | 16-bit SAR, SPI peripheral, VDD 2.7-3.6 V, external VREF pin 0.1 V..VDD, input range includes 0 V, offset AND gain specified as MAXIMA (**ADS8326IB** class) | VDD=**‘VDD_ADC‘** (behind R7), GND=‘GND‘, VREF=‘VREF‘, ‘+IN‘=‘/AIN_ADC‘, **‘-IN‘=‘/AGND_SENSE‘ (its OWN net: ‘U1.3‘ + ‘R5.2‘ + ‘R8.1‘) - read the note below**, CS=‘/CS‘, DCLOCK=‘/SCLK‘, DOUT=‘/DOUT‘ |
| ‘U2‘ | 2.048 V series voltage reference, <= 0.02 % initial max, <= 2 ppm/degC max, Vin_min <= 3.035 V (**ADR4520B** class) | IN=‘+3V3‘, GND=‘GND‘, OUT=‘VREF‘ |
| ‘U3‘ | unity-gain follower, CMOS input, RRIO, Vos <= 100 uV max, Ib <= 400 pA over temp (**OPA320** FITTED - chosen on settling from the converter’s clock-count acquisition window; **OPA333** the same-footprint alternate) | V+=‘+3V3‘, V-=‘GND‘, ‘+IN‘=‘/AIN_DIV‘, ‘-IN‘=‘/AIN_BUF‘, OUT=‘/AIN_BUF‘ |
| ‘R1‘ ‘R2‘ ‘R3‘ | attenuator TOP arm, 3 x 200 kohm 0.02 % / 10 ppm, ONE part number with ‘R4‘/‘R5‘ | ‘/AIN_RAW‘-‘/ATT_A‘-‘/ATT_B‘-‘/AIN_DIV‘ |
| ‘R4‘ ‘R5‘ | attenuator BOTTOM arm, 2 x the same 200 kohm part | ‘/AIN_DIV‘-‘/ATT_C‘-‘GND‘ |
| ‘R6‘ | ADC input isolation, 20-100 ohm (sim benches 2/3 set it) | ‘/AIN_BUF‘ / ‘/AIN_ADC‘ |
| ‘R7‘ | **converter rail-entry isolation, 5-10 ohm (10 ohm class), 0402/0603 thick film** - per the ADS8326’s own recommended application circuit | ‘+3V3‘ / ‘VDD_ADC‘ |
| ‘C1‘ | 10 uF X7R bulk, **at the J2 entry, not at the converter** | ‘+3V3‘ / ‘GND‘ |
| ‘C2‘ | 100 nF X7R, converter VDD - **the SMALLER of the pair, so it sits closest to the pin** | ‘VDD_ADC‘ / ‘GND‘ |
| ‘C3‘ | converter VREF bypass - **47 uF class low-ESR** per the ADS8326’s reference-input section, which is inside the ADR4520’s 1-100 uF stable window (see ‘C5‘) | ‘VREF‘ / ‘GND‘ |
| ‘C4‘ | 100 nF X7R, reference IN | ‘+3V3‘ / ‘GND‘ |
| ‘C5‘ | reference OUT cap - **MANDATORY and >= 1 uF.** NOT a DNP candidate | ‘VREF‘ / ‘GND‘ |
| ‘C6‘ | 100 nF X7R, buffer V+ | ‘+3V3‘ / ‘GND‘ |
| ‘C7‘ | 1-2.2 nF **C0G/NP0** at the converter input pin | ‘/AIN_ADC‘ / ‘GND‘ |
| ‘C8‘ | **10 uF X7R at the converter VDD pin**, beside ‘C2‘ and behind ‘R7‘ | ‘VDD_ADC‘ / ‘GND‘ |
| ‘J1‘ | 2-pole 5.08 mm THT screw terminal, silk-marked SIG and GND | 1=‘/AIN_RAW‘, 2=‘GND‘ |
| ‘J2‘ | 6-pin 2.54 mm single-row male THT header | 1=‘+3V3‘, 2=‘GND‘, 3=‘/CS‘, 4=‘/SCLK‘, 5=‘/DOUT‘, 6=‘GND‘ |
| ‘H1‘-‘H4‘ | M3 clearance holes, 3.2 mm | mechanical (no net) |

**`U1 -IN` IS A DEDICATED SENSE RUN, NOT A GROUND SYMBOL** UPDATED at P4 review (E1). `-IN` is now its OWN NET, `/AGND_SENSE`, with exactly three nodes: `U1.3`, `R5.2` and `R8.1`. `R8` is a single 0 ohm tie to `GND` at the string bottom, and it is the ONLY join between them.

This changed because the earlier arrangement - `-IN` on `GND`, with the intent carried by placement - does not survive the pour. On the netlist `U1.3` was one of 16 `GND` nodes, so the B.Cu pour

thermal-connects pin 3 exactly like pin 4 and references the measurement to the pour AT THE CONVERTER, deleting the cancellation with every gate green. A corridor is a keep-clear swath, not a connectivity rule. Giving the sense its own net makes it netlist-enforced: the pour cannot cover `/AGND_SENSE`, the router must route `U1.3` to `R5.2`, and every gate checks it.

The reason: **a divider passes a ground offset at UNITY while dividing the signal by K = 0.400**, so any offset between the string's bottom node and the converter's negative reference is referred to the input multiplied by 1/K = 2.5. **1 mV of copper offset = 2.5 mV at the terminal**, half the 25 degC budget. Sensing at the string bottom cancels it exactly: `+IN` carries `V_tap + (GND_A - GND_C)`, `-IN` carries `(GND_A - GND_C)`, difference `V_tap`. **Bounded:** the ADS8326 specifies `-IN` at **-0.3 V to +0.5 V** relative to device ground, so the sense point must be a node whose offset stays in the millivolts - which R5's pad is, and which a distant or shared-return ground is not. The bottom of the string is a *reference tie*, not a ground connection: it must not share copper with a return carrying other current. Carried in `constraints.json` as the `R5->U1` corridor; see `blocks.md` s8.1.

**`R7` + `C2` + `C8`: the converter's rail entry, and it is datasheet-required** `R7` (5-10 ohm) goes between `+3V3` and `U1`'s VDD pin, **upstream of both caps**, with `C2` (0.1 uF, smaller, closest to the pin) and `C8` (10 uF) on the `VDD_ADC` side. This is the ADS8326's own recommended application circuit, so it is what the datasheet REQUIRES, not "filtering the datasheet does not require" - the distinction the scope tier actually draws. A SAR has no supply rejection at the instant that matters, because the spikes that hurt land just before the comparator latches.

**It isolates 'U1' alone.** `U2` and `U3` stay on `+3V3` upstream of `R7`, for the same reason the source figures put the reference and the op amps on the analog rail rather than behind the converter's RC. Cost: ~0.6 mV of DC drop at a ~60 uA typical draw, which costs the error budget nothing (the conversion is referenced to `VREF`, not to VDD) and leaves VDD inside 2.7-3.6 V on the worst -case rail. **No ferrite anywhere** - that one is still the reflex addition nothing here requires.

**`C5` is mandatory, and it is why there is no reference buffer on the BOM** The ADR4520 specifies a stable output-capacitance **WINDOW of 1 uF min to 100 uF max**, two-ended, and the 2.048 V member is one of the two that need the full 1 uF (the >= 3.0 V members need only 0.1 uF). The output cap is a compensation element inside the loop, so **both ends bind and `C5` cannot be DNP**. That same window contains the ADS8326's 47 uF-class REF bypass (`C3`), which is what closed the open risk that this board might need a reference buffer or a series isolation resistor between the reference and `C3`: **neither is fitted, and the reference drives the converter's bypass directly.** If a current-limiting resistor were ever needed there it goes between the reference SOURCE and `C3` - never between `C3` and the VREF pin, where it would sit in the per-bit pulse path.

**J2 is 6 pins, and the sixth is GND, not a spare.** The chosen converter's SPI is read-only (CS, DCLOCK, DOUT - no MOSI), so the block needs 5 pins. The answered Q6 default reserved 6. Spending the sixth on a second GND puts a return reference at BOTH ends of the digital group, which is the return-path requirement D3/D4 already impose, rather than leaving a floating pin. It is support for the interface, not a feature.

**No series damping on '/SCLK' or '/CS', no pull-up on '/CS'.** Both are "conditioning the datasheet does not require" and the scope tier excludes them; `/CS` is driven by the host at all times the board is powered, and the one hot-plug hazard that a pull-up would not fix is already recorded and accepted (answered Q5).

**The follower's feedback is a WIRE.** `U3 -IN` ties directly to `U3 OUT`, no resistor. A gain-setting network there would put two more tolerances and two more leakage nodes into the budget for no benefit - the attenuation is already done, precisely, upstream.

**4. Nets deliberately left OUT of `constraints.json`**

`/AIN_RAW` is declared only in `voltages` (5 V, which makes `check_creepage` a clean, visible no-op). `/ATT_A`, `/ATT_B`, `/AIN_DIV`, `/ATT_C`, `/AIN_BUF` and `/AIN_ADC` are declared nowhere, on purpose:

- A `power` entry on any of them would give a high-impedance sense node a minimum-width rule and pull it into `check_pdn`'s decoupling inventory. `/AIN_DIV` in particular wants to be SHORT and THIN and nowhere near anything, which is the opposite of what a width rule expresses.
- A `high_speed` entry would be a lie: they are DC nodes. Their sensitivity is to LEAKAGE and to CAPACITIVE COUPLING FROM the SPI nets, which is why the three SPI nets ARE declared `high_speed` (that check verifies THEIR return path is continuous, which is the mechanism that keeps their return current out of the analog region) and why the protections for the analog nodes live in `placement.corridors`, `placement.separation` and the prose rules D1-D5.

The rules that DO apply to them are layout rules, carried in `constraints.json` `placement` and in s5 below.

**5. Placement groups this sheet implies (they become P6 groups)**

- **'attenuator'** (anchor `U3`; `R1-R5`, `R6`, `C6`, `C7`): the five string elements in physical order R1->R5 with the R3/R4 junction adjacent to `U3`'s `+IN` pad. `/AIN_DIV` is the SHORTEST high-impedance run on the board - a few millimetres, not the most convenient route. **The guard ring is part of this group's geometry:** `/AIN_BUF` copper encircles the `/AIN_DIV` trace and `U3` `+IN` on F.Cu with clearance on both sides, and solder-mask relief over the ring. An annealer optimising a cost function cannot discover a guard ring; this cluster is placed by explicit `place_edit` and LOCKED before `place_anneal` runs.
- **'reference'** (anchor `U2`; `C4`, `C5`): sits beside `U1`'s VREF pin.
- **'converter'** (anchor `U1`; `C2`, `C3`, `C8`, `R7`): `C3` within 2.5 mm of the VREF pin, SAME layer, no via between pad and cap - and what binds is the charge LOOP (`C3` -> REF pin -> the internal array -> the converter's GND pin -> back), so wide copper out AND back, and nothing in series between cap and pin. `C2` within 2 mm of VDD with its ground via within 1 mm of its own pad - that loop is the SPI drivers' current loop and must stay under 20 mm^2 (D3). `R7` upstream of `C2`/`C8`, `C2` (the smaller) closest to the VDD pin.
- **'entry'** (anchor `J2`; `C1`): the bulk cap at the rail entry, not at the converter.

Spatial order across the board, from D4: **`J1` -> `attenuator` -> `converter` + `reference` -> `J2`**, with `J1` and `J2` on DIFFERENT edges, openings outward. Everything is on the FRONT side - single-sided assembly, and a part moved to the back would also cut the B.Cu return pour that D1 requires unbroken, so every ref carries an explicit `sides: front`.

**6. Handover to P4**

1. Build the root sheet flat. Power symbols for `+3V3`, `GND`, `VREF` (the last one with its Value edited); root local labels for everything else. 2. `netlist_audit.py --net ... --constraints ...` must come back clean before P5 touches the board. 3. Any net whose exported name differs from s2 is reconciled IN `constraints.json` at that moment, not later - an entry naming a net that does not exist fails silently, which is the whole reason this board is flat.

Full architecture narrative (not inlined; see the artifact index): `architecture/blocks.md`, `architecture/power_tree.md`, `architecture/stackup.md`.

# 5 Schematic

Gate `erc` (phase P4): **pass** after 4 attempt(s); last 2026-08-20T00:46:04, failing 0 of 0.

### Review waivers

### ERC / schematic-review waivers - bb-adc

Machine ERC is 0/0 (gate `erc` PASS, commit 1befa1e). This file records the schematic-review WARNINGS that are accepted rather than fixed, and why. The single ERROR (E1, the Kelvin sense not surviving the pour) was FIXED, not waived.

#### netlist_audit: `power_no_consumers` on VREF - ACCEPTED

U1's REF pin is typed `passive` in the pulled symbol, so a declared power net feeds no `power_in` pin. The netlist is correct and nothing downstream keys off pin type for this net (`check_current` reads the net name). NOT hand-fixed on purpose: `lib_pin_types.py` has no NC concept and was dry-run-confirmed to silently revert U2's five `no_connect` pins, so a manual retype becomes a standing re-apply obligation for a cosmetic warning.

#### W1 reference headroom vs rail ripple - ACCEPTED AS ANALYSIS

The 87 mV DC margin was compared against DC tolerance alone, while the verified record says tolerance + IR drop + ripple trough, and the owner's answered rail carries "tens of mV" of noise. Accepted because the output capacitance answers it: ~49 uF against a microamp load holds VOUT through input dips of microseconds, and a series-reference dropout spec is a DC condition. No input RC is added - the scope tier excludes filtering the datasheet does not require, and this analysis says it is not required. BINDING CONSEQUENCE, now stated: the design is validated for a host rail whose DC level stays at or above 3.135 V. Below that the margin is gone and no capacitor recovers it.

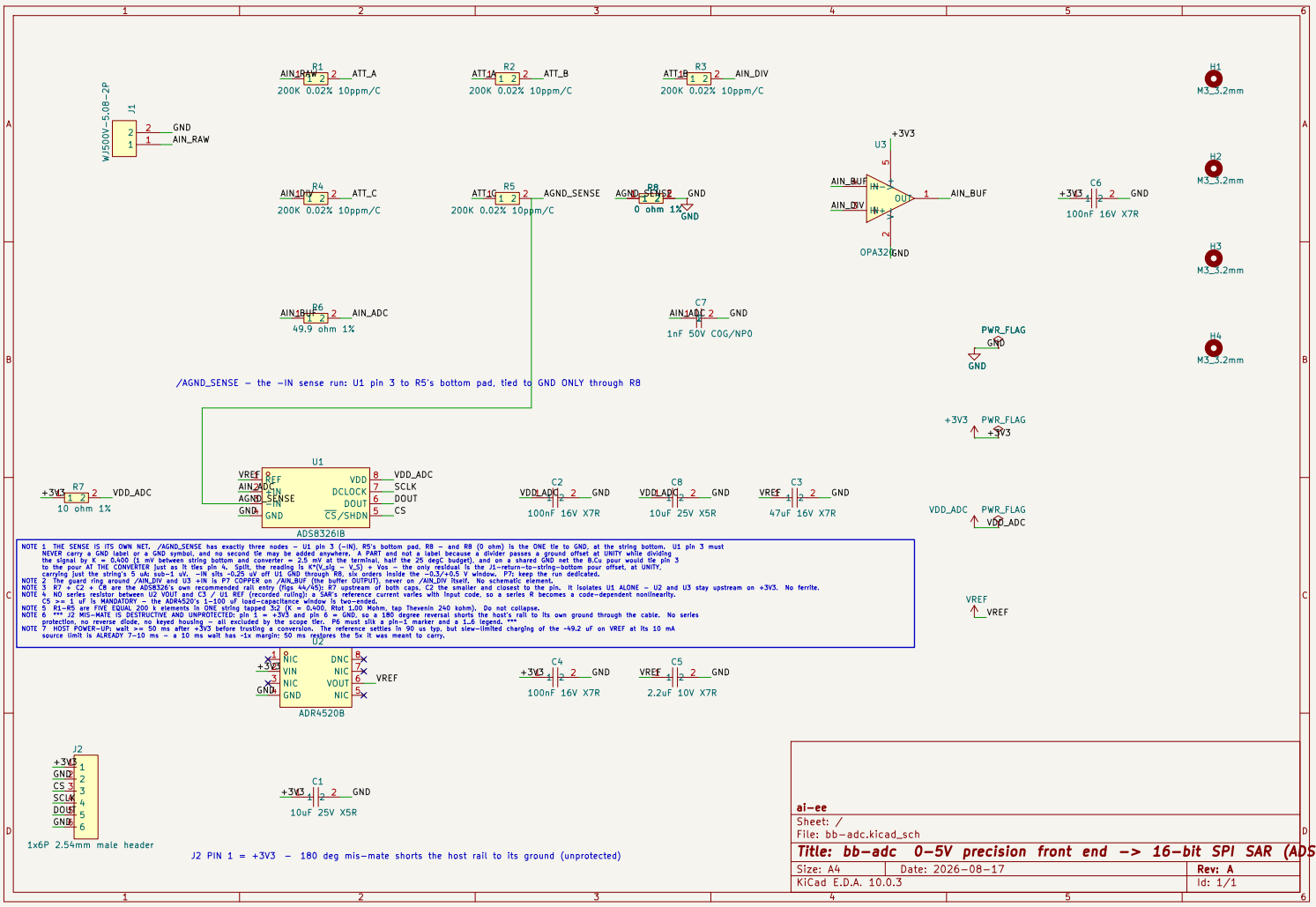
#### W5 C7 guaranteed-minimum capacitance - DEFERRED TO THE P8 BENCH

The 20x C_SH floor (960 pF for a 48 pF sampling cap) was checked on NOMINAL 1 nF; the fitted J-grade part guarantees only 950 pF. Not fixed now because R6/C7 are explicitly provisional pending `acquisition-settling` and `buffer-stability`. The bench must choose C7 on its GUARANTEED minimum, with R6 present - the isolation resistor is what lifts the pure-capacitive-load limit that would otherwise cap C7 at 1 nF.

#### Not waivers - fixed or actioned elsewhere

- E1 Kelvin sense: FIXED (own net `/AGND_SENSE` + a single 0 ohm tie R8).
- W2 error budget computed on the wrong amplifier: RESTATED for the OPA320 (RSS 3.25 / 3.42 mV, worst case 8.58 / 10.74 mV).
- W3 host power-up wait: RAISED 10 ms -> 50 ms.
- W4 J2 mis-mate is destructive: STATED on the sheet, on the silkscreen at P6, and in the design doc. No protection hardware (excluded by scope; the checklist asks for a statement).
- constraints.json VREF current 7 uA -> 14 uA max per the datasheet row. Nothing moves.

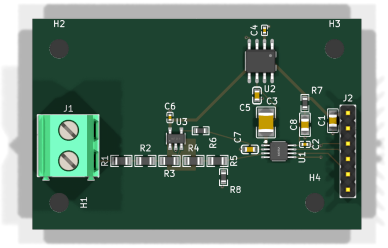
The full schematic PDF follows.


R1
AIN1RAW
ATT_A
200K 0.02% 10ppm/C
R2
ATT_B
200K 0.02% 10ppm/C
R3
AIN_DIV
200K 0.02% 10ppm/C
H1
M3_3.2mm
J1
WJ500V-5.08-2P
GND
AIN_RAW
H2
M3_3.2mm
R4
AIN1DIV
ATT_C
200K 0.02% 10ppm/C
R5
AGND_SENSE
R8
0 ohm 1%
GND
U3
+3V3
AIN_BUF
AIN_DIV
OUT
OPA320
C6
100nF 16V X7R
H3
M3_3.2mm
R6
AIN1BUF
AIN_ADC
49.9 ohm 1%
C7
AIN1ADC
1nF 50V C0G/NP0
PWR_FLAG
H4
M3_3.2mm
/AGND_SENSE — the −IN sense run: U1 pin 3 to R5's bottom pad, tied to GND ONLY through R8
R7
10 ohm 1%
VDD_ADC
U1
VREF
AIN+
AGND_SENSE
AIN−
GND
VDD
DCLOCK
DOUT
CS/SHDN
SCLK
DOUT
CS
ADS8326IB
C2
100nF 16V X7R
C8
10uF 25V X5R
C3
47uF 16V X7R
NOTE 1 THE SENSE IS ITS OWN NET. /AGND_SENSE has exactly three nodes — U1 pin 3 (−IN), R5's bottom pad, R8 — and R8 (0 ohm) is the ONE tie to GND, at the string bottom. U1 pin 3 must NEVER carry a GND label or a GND symbol, and no second tie may be added anywhere. A PART and not a label because a divider passes a ground offset at UNITY while dividing the signal by K = 0.400 (1 mV between string bottom and converter = 2.5 mV at the terminal, half the 25 degC budget), and on a shared GND net the B.Cu pour would tie pin 3 to the pour AT THE CONVERTER just as it ties pin 4. Split, the reading is K*(V_sig − V_S) + Vos — the only residual is the J1-return-to-string-bottom pour offset, at UNITY, carrying just the string's 5 uA; sub-1 uV, −IN sits +0.25 uV off U1 GND through R8, six orders inside the −0.3/+0.5 V window. P7: keep the run dedicated.
NOTE 2 The guard ring around /AIN_DIV and U3 +IN is P7 COPPER on /AIN_BUF (the buffer OUTPUT), never on /AIN_DIV itself. No schematic element.
NOTE 3 R7 + C2 + C8 are the ADS8326's own recommended rail entry (figs 44/45): R7 upstream of both caps, C2 the smaller and closest to the pin. It isolates U1 ALONE — U2 and U3 stay upstream on +3V3. No ferrite.
NOTE 4 NO series resistor between U2 VOUT and C3 / U1 REF (recorded ruling): a SAR's reference current varies with input code, so a series R becomes a code-dependent nonlinearity. C5 >= 1 uF is MANDATORY — the ADR4520's 1–100 uF load-capacitance window is two-ended.
NOTE 5 R1–R5 are FIVE EQUAL 200 k elements in ONE string tapped 5:2 (K = 0.400, Rtot 1.00 Mohm, tap Thevenin 240 kohm). Do not collapse.
NOTE 6 *** J2 MIS-MATE IS DESTRUCTIVE AND UNPROTECTED: pin 1 = +3V3 and pin 6 = GND, so a 180 degree reversal shorts the host's rail to its own ground through the cable. No series protection, no reverse diode, no keyed housing — all excluded by the scope tier. P6 must silk a pin-1 marker and a 1.6 legend. ***
NOTE 7 HOST POWER-UP: wait >= 50 ms after +3V3 before trusting a conversion. The reference settles in 90 us typ, but slew-limited charging of the ~49.2 uF on VREF at its 10 mA source limit is ALREADY 7–10 ms — a 10 ms wait has ~1x margin; 50 ms restores the 5x it was meant to carry.
U2
NIC
VIN
NIC
GND
DNC
NIC
VOUT
NIC
VREF
ADR4520B
C4
100nF 16V X7R
C5
2.2uF 10V X7R
J2
+3V3
GND
CS
SCLK
DOUT
GND
1x6P 2.54mm male header
C1
10uF 25V X5R
J2 PIN 1 = +3V3 — 180 deg mis-mate shorts the host rail to its ground (unprotected)
ai-ee
Sheet: /
File: bb-adc.kicad_sch
Title: bb-adc 0–5V precision front end -> 16-bit SPI SAR (ADS8326I
Size: A4
Date: 2026-08-17
Rev: A
KiCad E.D.A. 10.0.3
Id: 1/1

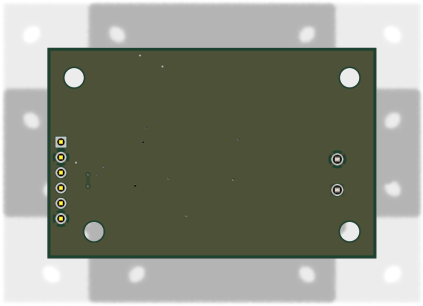
# 6 Layout

Gate `place` (phase P6): **pass** after 7 attempt(s); last 2026-08-19T04:57:37, failing 0 of 0.

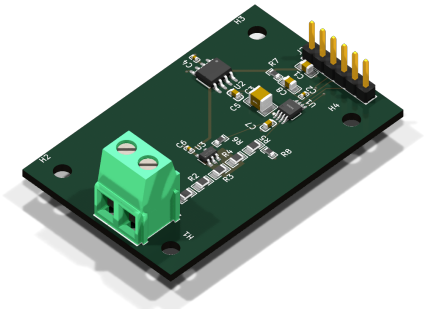
Gate `drc_routed` (phase P7): **pass** after 11 attempt(s); last 2026-08-20T00:46:26, failing 0 of 0.



`reports/renders/bb-adc_top.png`



`reports/renders/bb-adc_bottom.png`



`reports/renders/bb-adc_iso.png`

# 7 Verification

Gate `verify` (phase P8): **pass** after 9 attempt(s); last 2026-08-19T04:53:50, failing 0 of 1. *reports/verify_all.json not available.*

### Design review of record

### review-board - bb-adc, P8 adversarial board review

Fresh-context review of `kicad/bb-adc.kicad_pcb` (54.75 x 34.92 mm, 2L, 1 oz, JLC2313_1.6, 25 footprints, 21 nets). Gates going in: `erc` PASS, `drc_routed` PASS 0/0, `sim` PASS, `verify` PASS with one waiver.

Build mode `learning block-basics:` -> scope tier **block-only**, binding **canonical**. Absent protection, ESD, indicators, test points, second rails and filtering the datasheet does not require are OUT OF SCOPE and are not reported below. The outline was judged against the 54.75 x 34.92

the placement EARNED (`reports/board_edit-fit.json`, from a 63.95 x 66.03 provisional), never against a stated number.

Renders made for this review: `reports/renders/bb-adc_top.png`, `bb-adc_bottom.png`, `bb-adc_iso.png`.

**5 errors, 5 warnings, 2 waivers recommended.** (W2 below is a waiver verdict, not a finding, so it carries no entry in `review-board.json`.)

—

### E1 - J1's wire entry faces INTO the board. The terminal is 180 degrees out.

`reports/renders/bb-adc_top.png` - the two dark throats in the green terminal body are on its EAST face, pointing at R1. The west face, 1.0 mm from the board edge, is a closed wall.

Proved from the library, not from the render. In `lib/aiee.pretty/CONN-TH_2P-P5.00_WJ500V-5.08-2P.kicad_mod` the pads sit at local `(+/-2.54, 0)` and the body spans local y `-5.64 .. +4.52`. The vendor 3D model (`aiee.3dshapes/...wrl`, VRML y = -footprint y) has two openings 3.0 mm wide spanning z 3.0 .. 9.8 mm, centred on x = +/-2.54 - i.e. directly in line with the two pins - in the face at footprint y = **+4.50**. The opposite face (y = -5.50) is solid except for ribs at z 0..1.0 and 6.5..6.69. The wire entry is therefore the face NEARER the pins, which is also how this class of block is built.

J1 is placed at (24.500, 39.060) rot 90, which maps footprint +y to board +x. The throats open EAST at x = 29.0, 2.2 mm from R1's pad and 11 mm from the west edge the body is parked against.

Three artifacts say this is wrong:

- `kicad/constraints.json` `placement.edges` J1: *"Screw terminal, opening outward."*
- `requirements.md` s5: *"the screw terminal and the host header sit on different board edges with their openings facing outward, so the analog input wiring does not have to cross the digital header to reach the bench."*
- the schematic's J1 note: *"ANALOG IN 0-5V: 1=SIG (Rs<=200ohm), 2=GND. LEFT edge, opening outward"*.

Consequence: the field lead must be inserted from the board's interior and then laid back across the 1 Mohm divider string, the buffer and the J2 ribbon to reach the bench - the exact geometry the constraint exists to forbid, and a metre of unscreened wire draped over a guarded 240 kohm node.

Fix: rotate J1 to rot -90. Pads keep x = 24.5; `/AIN_RAW` and `GND` swap to y = 36.520 / 41.600, so both need re-routing. The body then spans x 20.0..30.0

- 2.0 mm off the west edge and 1.2 mm from R1's pad, so a ~1 mm westward nudge

is worth taking with it. See also W6: the footprint's own silk arrows point at the closed wall.

### E2 - J2 carries no pin-1 mark that survives assembly, and the schematic makes one mandatory

The schematic says it twice, once in capitals with asterisks:

> NOTE 6 '*** J2 MIS-MATE IS DESTRUCTIVE AND UNPROTECTED: pin 1 = +3V3 and > pin 6 = GND, so a 180 degree reversal shorts the host's rail to its own > ground through the cable. No series protection, no reverse diode, no keyed > housing - all excluded by the scope tier. P6 must silk a pin-1 marker and a > 1..6 legend. ***'

The board's pin order confirms the hazard exactly: J2 = +3V3 (y 33.650), GND (36.190), /CS (38.730), /SCLK (41.270), /DOUT (43.810), GND (46.350). Reversed, the host's 3V3 lands on board pin 6 = GND.

What is actually on the silk: one line at y = 34.920 dividing the pin-1 cell, and an `fp_circle` at (69.250, 32.380) of radius **0.03 mm** stroked 0.15 - a 0.21 mm blob sitting exactly on the corner where two 0.25 mm outline strokes meet, so it is swallowed by the outline. Both live inside the silk rectangle x 69.250..71.750 by y 32.380..47.620 = 2.50 x 15.24 mm, which is *under* the header's 2.54 x 15.24 mm plastic body. There is no "1", no 1..6 legend and no rail names anywhere.

After assembly nothing is readable from the mating side. `bb-adc_top.png` shows the black body filling a plain white rectangle. The only surviving indicator is pad 1's square shape, visible only from the bottom (`bb-adc_bottom.png`).

Fix: silk text west of x = 69.25 - there is clear board from x 66.4 to 69.25 for most of the run, and the C1/C2 refdes can move. A "1" plus the six names is what the schematic asked for.

### E3 - J1 has no SIG/GND marking, and the design doc says it does

`reports/design_doc/bb-adc-design-doc.tex` publishes:

> `J1` | 2-pole 5.08 mm THT screw terminal, **silk-marked SIG and GND** | > 1=`/AIN_RAW`, 2=`GND`

`requirements.md` s2 states the same requirement, and s8 makes it load-bearing: *"the board has no input protection by mode, so the silk marking and the owner's answer to Open question 5 are the only defenses against a wrong connection."*

The board's silkscreen contains 25 texts. All 25 are reference designators. There is no SIG, no GND, no polarity legend. J1's pin-1 dot (`fp_circle` at (29.000, 44.270), radius 0.03) sits on the body corner and is covered by the 10.0 x 10.76 mm plastic body, so it does not survive assembly either.

A published claim that the artifact does not carry is a defect in both.

### E4 - the contracted input-impedance numbers are missing from the silkscreen

The P2 ruling that narrowed the owner's answered Q9 (`requirements.md` s9, `architecture/blocks.md`, and the design doc's decision log) reads:

> the contract is now **source <= 200 ohm** ... and **board presents > 1.00 Mohm **...** Both numbers go on the silkscreen.** ... the contract > belongs on the silk because input impedance is a specification of the > instrument

Neither number appears on F.SilkS or B.SilkS. This one matters more than ordinary legend text: the ruling NARROWED what may legally be connected to the terminal by 5x, and the silk was named as the place that travels with the board. A 1 kohm source - inside the envelope the owner originally answered - now produces 5 mV of loading error, the entire 25 C budget, with nothing on the hardware to say so.

### E5 - GND copper sits inside the closed guard ring

**First, the closure, re-proved geometrically and independently of any render - the ring IS closed, with margin.**

1. Polygon evidence. The `/AIN_BUF` zone's filled polygon on F.Cu is a single island whose ring has exactly **one interior hole**, area 7.951 mm2, bounds (41.106, 38.523) - (44.494, 42.416). Zone copper 17.176 mm2; 19.08 mm2 with the net's own pads and the U3.1/U3.4 stub, matching the 19.03 mm2 claimed. 2. Raster evidence. Rasterised at 2.5 um over an 8.2 x 8.8 mm box, a flood fill started from every `/AIN_DIV` cell and allowed to spread through *all* non-`/AIN_BUF` space, 8-connected so it can leak diagonally (the conservative direction), never reaches the box border. 3. Margin. Eroding the guard copper by 200 um on every edge (19.11 -> 10.18 mm2 of pour) leaves the ring still closed. Closure in the as-built copper is not marginal; the "does not close at 0.2 mm" figure describes re-pouring at a larger zone clearance, not the copper on this board. 4. Drive. The zone is on `/AIN_BUF` and is galvanically continuous with U3 pin 1 (OUT) and pin 4 (-IN) - same potential, low impedance, as record `resistive-attenuator-high-z-tap-guard-and-leakage` requires. Clearance to the node

is a uniform 0.1275 mm all the way round. The 19.08 mm2 plate adds ˜0.5 pF to the follower output over the 1.53 mm core; no stability cost, and R6 = 49.9 ohm isolates the 1 nF bucket regardless.

**The defect is what the closed ring encloses along with the node.** Inside the guarded enclosure there are two GND conductors: U3 pin 2 (V-) and a 0.6 mm GND via at (41.850, 39.850). Measured F.Cu surface gaps at the tap:

```
| path | gap |
|---|---|
| ‘/AIN_DIV‘ trace -> GND via (41.850, 39.850) | **0.227 mm** |
| ‘/AIN_DIV‘ trace -> U3.2 pad | 0.289 mm |
| U3.3 pad -> U3.2 pad | 0.350 mm |
| ‘/AIN_DIV‘ -> nearest guard copper | 0.1275 mm (but on the *outside* of all
three) |
```

The guard intercepts nothing on those three paths, and they carry the largest potential difference anywhere near the node.

Priced with `architecture/blocks.md` s3.6's *own* model - a 1 Gohm surface path under no-clean flux plus humidity, "and JLC does not wash boards":

- the +3V3 path the guard was built to stop drives 3.3 - 2.0 = 1.3 V -> 1.3 nA -> 1.3 nA x 240 kohm / 0.400 = **0.78 mV** at the terminal (the document's own number);
- the unguarded GND path drives the full node voltage, 2.000 V at a 5.000 V reading -> 2.0 nA -> **1.20 mV** at the terminal.

Error-budget row 9, "Divider-node leakage, GUARDED (estimate)", carries **0.10 mV**. It is understated about 12x for the path that now dominates, and row 9 is the table's only +/-3x estimate. RSS at 25 C moves 3.23 -> 3.44 mV (still inside the 5.00 mV target) and the worst-case sum moves up ˜1.1 mV, so the board still closes its spec - but the guard's stated 3.3e5 collapse factor does not hold for the leakage term that matters, and a 0.1 %-class instrument publishing 3.25 mV RSS should not rest on a number that is wrong by an order of magnitude.

Split of the fix:

- The GND via at (41.850, 39.850) is a free routing choice. It sits 0.65 mm south of U3.2, i.e. on the pin-3 side. Moving it north or west of the pad, or outside the ring entirely, restores 0.227 -> >0.5 mm. **This is the part to fix.**
- The U3 pin2/pin3 0.350 mm channel is irreducible in SOT-23-5 at the fab floor: a guard trace needs 0.127 + 2 x 0.127 = 0.381 mm and only 0.350 mm exists. It has to be re-stated in row 9, not fixed in copper. (Also note that a stated purpose of the guard was to keep the board's dominant uncertainty on vendor maxima rather than on a leakage estimate - that argument is weakened by exactly this much.)

Not raised as drift, for the record: the guard is single-sided where the record asks for "both sides of the board". That is an explicit recorded trade in `constraints.json` corridors R3->U3 ("here the back face is the unbroken B.Cu GND pour that D1 requires, so the ring is single-sided by a recorded trade"). B.Cu under the node measures 100 % GND pour. The trade is legitimate; what it never records is the record's own price for it - bulk leakage falls only about one order of magnitude, set by board thickness and ring width.

—

### W1 - the one verify_all WARNING is a false positive, and its cause is a real hole

`check_pdn` / `pdn_no_bulk`: *"power rail +3V3 has 2 cap(s) but no bulk reservoir (>= 1 uF); only ceramics hold it"*, refs C4, C6.

Wrong on the facts. C1 = 10 uF X5R 0805 sits on +3V3/GND at (68.000, 34.920), 0.925 mm (pad edge) from J2 pin 1 - precisely the "bulk 10 uF at the RAIL ENTRY" that `constraints.json`'s `entry` group specifies and earns by arithmetic. The rail has its reservoir.

The cause is that `kicad/decoupling.json` lists six associations - C2, C3, C4, C5, C6, C8 - and **omits C1**. C7 is correctly absent (signal-path charge bucket, not decoupling). So `check_pdn` counted 0.2 uF, and, more to the point, `check_decoupling` never examined the rail-entry bulk at all: it passed by not being asked.

**Verdict: waive the warning, fix the metadata.** Add '{"cap": "C1", "ic": "J2", "pin": "1", "rail": "+3V3", "value": "10uF 25V X5R"}' and re-run so the bulk is actually checked rather than assumed.

### W2 - the /CS return-path waiver: I agree, and here is the independent check

Both of the waiver's reasons hold up.

*(a) Out of model* - verified. F.Cu carries exactly one zone, `/AIN_BUF` (19.08 mm2, 12+ mm west of the crossing). There is no GND zone on F.Cu at all, so `check_return_path`'s adjacent-layer model gives every B.Cu trace zero reference copper by construction. The reported "0.03 mm / 0.20 mm2" is what survives of the 1.93 mm tunnel after the checker's own excision disks around the two /CS vias (0.3 mm via radius + 0.5 mm zone clearance = 0.8 mm each). It is a bookkeeping remainder, not a measurement of anything physical.

*(b) Topologically forced* - verified. At U1 the order down the page is /SCLK (y 39.680), /DOUT (40.330), /CS (40.980); at J2 it is /CS (38.730), /SCLK (41.270), /DOUT (43.810). /CS must cross both other signals: a 3-cycle, and one layer cannot resolve it.

*(c)* Refusing to shorten the tunnel 0.09 mm so the excision disks overlap, at the cost of ~0.04 mm of via-to-trace DRC margin, is the right call. Spending manufacturing margin to land inside a checker's arithmetic is a bad trade.

**Uphold the waiver.** One thing to name that the checker did *not* report, on the other side of the same geometry: the B.Cu tunnel plus its two via antipads cut a ~1.2 mm-wide slot in the GND pour, and both `/SCLK` (y 39.680) and `/DOUT` (y 40.330) cross it on F.Cu at x ~ 66. Each crossing happens to fall inside a /CS via's excision disk, so neither surfaced. At 5 ns edges the added return inductance is ~1 nH and the site is 12+ mm east of the analog section, so I do not escalate it - but it is a second reason the next-revision fix the waiver already names (J2 pin order CS, DOUT, SCLK) is worth doing: it removes the slot as well as the bookkeeping.

### W3 - 87 mV of reference headroom, with the rail noise not yet subtracted

`blocks.md` s8.2 records the number honestly: the ADR4520's guaranteed input floor is 3.048 V (VOUT 2.048 + VDO_max 1.000 at no load and at 2 mA, record `series-voltage-reference-input-headroom-gate`), against the 3.135 V worst-case rail - **87 mV**.

The layout does not eat it: J2 pin 1 to U2 is ~18.9 mm of 0.5 mm 1 oz copper = ~19 mohm, and at the 2.7 mA peak draw that is 0.05 mV. Good.

What is not in the recorded analysis is the rail spec the owner answered in Q7: "tens of millivolts of noise". A 40-50 mV negative excursion eats half to all of the margin, and below the floor VOUT has *already* degraded 0.1 % = 2 mV, which maps one-for-one to gain error = 5 mV at the terminal (`sar-adc-reference-sets-the-gain`). C1 (10 uF) + C4 (0.1 uF) + C5 (2.2 uF on the output, holding a 14 uA load) absorb a short dip; a sustained low rail they do not. None of the six sim benches covers input headroom.

Recommend a bring-up line rather than a board change: measure VIN at U2 pin 2 and require >= 3.05 V under the host's worst rail.

### W4 - the mounting holes are a trapezoid, not a rectangle

H1 (22.450, 48.500), H2 (22.450, 23.000), H3 (68.300, 23.000), H4 (65.000, 48.500). H4 is 7.75 mm in from the east edge where H3 is 4.45 mm.

Forced, and I can see why: H4 at (68.3, 48.5) would drive its 3.6 mm courtyard into J2's (x 69.250..71.750, y 32.380..47.620) on a 34.92 mm-tall board whose 15.24 mm header is centred at y = 40. `requirements.md` s5 asked for "4 x M3 clearance holes (3.2 mm) inset from the corners" and explicitly allows two on opposite corners if the outline is too small, so this is inside the letter of the requirement and the board still mounts on four points.

Reported so the choice is visible: any bench plate or standoff set has to be drilled to a trapezoid, and nothing in the artifacts says so.

### W5 - requirements.md still describes a J2 the board improved on

`requirements.md` s2 / Q6 says "4 signal pins for SPI, so 6 pins for SPI". The board fits 6 pins as three signals plus a second ground: +3V3, GND, /CS, /SCLK, /DOUT, GND. That is better, and it is deliberate and recorded (`constraints.json` edges J2: "ground adjacent to the digital group at both ends of it"). The stale sentence is in the P0 artifact only. No board change - sync the document so the artifact set stops disagreeing with itself.

### W6 - the J1 footprint's silk insertion arrows are on the wrong face

`CONN-TH_2P-P5.00_WJ500V-5.08-2P.kicad_mod` draws two filled arrow polygons at local x -3.0..-2.0 and +2.08..+3.08, running from local y -5.5 to -2.5 and pointing inward - i.e. on the y = -5.64 face. The wire throats are in the y = +4.50 face (E1). On the placed board the arrows land at board y ~ 41.56, x 19.0 -> 22.0, pointing the operator at the closed west wall.

Independent of how E1 is resolved, the footprint is internally inconsistent and misinforms whoever wires the terminal. Fix the library part, not just the placement. (The footprint name also says `P5.00` while the pads are correctly at 5.080 mm pitch - cosmetic, but it is the same file.)

—

### Verified clean

Recorded so a later reader knows these were checked, not skipped.

- **'/AGND_SENSE' is intact and nothing defeats it.** The net is exactly {U1.3, R5.2, R8.1}. Two F.Cu tracks (7.183 mm R5.2 -> U1.3, 1.950 mm R5.2 -> R8.1), **zero vias, zero B.Cu copper**, and there is no F.Cu GND zone anywhere on the board, so there is no pour contact to have. R8 is on the FRONT side at (49.800, 44.300). The sense tap is R5's own bottom pad, so the string's 5 uA return through R8 is outside the sensed node. Nearest GND copper is U1 pin 4 at 0.243 mm - VSSOP-8 0.65 mm pitch, unavoidable and harmless. The single-point tie holds.
- **Single-sided assembly holds.** 0 SMD pads on B.Cu, 0 flipped or back-side footprints. B.Cu carries the GND pour and one 1.93 mm /CS track. `bb-adc_bottom.png` shows THT pads and four holes only.
- **Placement separations from 'constraints.json' all pass with margin:** R1-R5/U3 vs J2 19.238 mm (>= 10), R1-R5 vs U1 5.676 mm (>= 5), U2/C5 vs J2 12.181 mm (>= 5), U2 vs H1/H2 29.771 mm and vs H3/H4 8.521 mm (>= 5). D2's 2.5 mm SPI-to-analog routing floor holds at 2.605 mm pad-to-pad (U1's own pins) and 3.911 mm trace-to-trace.
- **Decoupling geometry matches the records.** C2 (0.1 uF) 0.722 mm from U1 pin 8 with C8 (10 uF) behind it at 3.640 mm - "smaller capacitor closest to the pin" (`sar-adc-supply-bypass-and-rail-isolation`). C3 (47 uF) 1.291 mm edge / 2.290 mm centre from U1 pin 1, same layer, no via between pad and cap (spec <= 2.5 mm). C5 0.917 mm from U2 pin 6; C3 + C5 = 49.2 uF, inside the ADR4520's 1-100 uF window.

- **Outline earned, not given.** Edge.Cuts 18.00..72.75 x 18.03..52.95 = 54.75 x 34.92 mm, from the 63.95 x 66.03 provisional via `board_edit --outline fit` (`reports/board_edit-fit.json`). Judged against what the design earned.
- **No block or interface is missing.** B1 (R1-R5), B2 (U3 + C6), B3 (U1 + R6/ C7 + R7/C2/C8), B4 (U2 + C4/C5), C1 rail entry, R8 sense tie - all present. J1 2-pole 5.08 THT, J2 6-pin 2.54 single row, 4 x M3 3.2 mm all present. Power3V3 at 0.5 mm on +3V3 (44.07 mm of track) and the 0.6 mm VREF segment C3.1 -> U1.1 (2.290 mm) are both on the board as cited.
- **check_silk's blind spot, named.** It passed, and correctly: it checks silk-over-pad, legibility and refdes attribution. It has no concept of a *required* marking, nor of whether a marking survives assembly. E2/E3/E4 all live in that gap.

**Waivers recommended**

1. **Uphold** the existing `check_return_path` /CS waiver (W2). Both stated reasons verified independently; the residual is a checker artifact. 2. **Waive** the `check_pdn pdn_no_bulk` warning on +3V3 (W1) - the bulk exists - but treat the incomplete `decoupling.json` as a defect to fix, not as part of the waiver.

**Open - what I could not settle**

- E1's wire-entry direction rests on the vendor 3D model plus the pin-offset geometry, not on a vendor mechanical drawing I fetched. The two sources inside the footprint disagree with each other, so confirm against the WJ500V-5.08-2P drawing before rotating the part. Either way the footprint is self-contradictory and W6 stands.
- The 1.20 mV in E5 scales linearly with the real surface-insulation resistance, which is an estimate (blocks.md prices it +/-3x). A washed board makes it negligible; a humid, unwashed one makes it worse.
- Whether the trapezoidal hole pattern (W4) is a real nuisance depends on a bench plate I cannot see.

# 8 DFM and Fabrication

Gate `dfm` (phase P9): **pass** after 2 attempt(s); last 2026-08-19T04:56:31, failing 0 of 0. DFM findings by severity: none.

## Fabrication outputs

Exported layers: F.Cu, B.Cu, F.Silkscreen, B.Silkscreen, F.Mask, B.Mask, F.Paste, B.Paste, Edge.Cuts.

## Bill of materials

| Comment | Designator | Footprint | LCSC |
|---|---|---|---|
| 10uF 25V X5R | C1,C8 | C0805 | C15850 |
| 100nF 16V X7R | C2,C4,C6 | C0402 | C1525 |
| 47uF 16V X7R | C3 | C1210 | C5440143 |
| 2.2uF 10V X7R | C5 | C0603 | C513691 |
| 1nF 50V C0G/NP0 | C7 | C0603 | C106246 |
| WJ500V-5.08-2P | J1 | CONN-TH_2P-P5.00_WJ500V-5.08-2P | C8465 |

| Comment | Designator | Footprint | LCSC |
|---|---|---|---|
| 1x6P 2.54mm male header | J2 | HDR-TH_6P-P2.54-V-M-1 | C37208 |
| 200K 0.02% 10ppm/C | R1,R2,R3,R4,R5 | R0805 | C23067436 |
| 49.9 ohm 1% | R6 | R0603 | C23185 |
| 10 ohm 1% | R7 | R0603 | C22859 |
| 0 ohm 1% | R8 | R0603 | C21189 |
| ADS8326IB | U1 | VSSOP-8_L3.0-W3.0-P0.65-LS5.0-BL | C544731 |
| ADR4520B | U2 | SOIC-8_L5.0-W4.0-P1.27-LS6.0-BL | C579305 |
| OPA320 | U3 | SOT-23-5_L3.0-W1.7-P0.95-LS2.8-BR | C92494 |

2 CPL rotation correction(s) applied; no missing LCSC numbers. *fab/order.json not available.*

# 9 Run Record (Appendix)

## Phase digests

### P0

```
# P0 Intake - digest

- Mode ‘learning block-basics:‘ -> scope **block-only**, binding **canonical**,
stage none.  Geometry is an OUTPUT: no size stated, none invented, ‘--outline
auto‘.
- ‘requirements.md‘ written; ‘check_requirements.py‘ exit 0 / 0 violations,
mode leg live.
- Owner answered all 9 questions in one batch:  **0.1 % class** at the terminal
uncalibrated (+/-5 mV @25 C, +/-12 mV over 0-50 C), **SPI**, **3.3 V only on
J2**,
low-voltage bench source, Q4/5/6/7/9 at defaults.
- The problem this board teaches, fixed at P0:  0.1 % delivered THROUGH an
attenuator
that brings 0-5 V into a 3.3 V domain - ratio error and TCR are first-class
terms
beside the reference, not the converter’s bit count.
- Safety:  5 V max on any net; mains/battery/motor/>30 V/high-I/RF do not
apply, each
contingent on the Q1 envelope (no isolation, no protection, input GND = host
GND).
- Mode boundary for P1/P2:  block-only excludes a second rail the block does
not need
*to work*; if 0-5 V from 3.3 V honestly needs one, that is support, and a
decision.
- Owner then delegated design decisions ("decide yourself, best learning for
the basic
system"):  H1-H4 become report-and-proceed; only a safety change re-blocks.
- Run-level:  fable 5 out of credits -> every fable tier substitutes one step
up.
- Defect:  ‘modeslib.detect‘ skips ‘>‘-lines, so a blockquoted brief silently
disables
the mode leg of the lint.  Fixed here; ‘bb-mcu‘ has it too (reported, not
touched).
```

### P1

```
# P1 Research - digest
```

```
Roster:  2x component-scout (‘adc‘, ‘afe-support‘), reference-design (‘afe‘),
power-architect.  No ‘research-interface-spec‘:  SPI-into-an-unspecified-host
and a
screw terminal bind no standard.  Detail lives in research/*.md.

- Topology = attenuator -> low-Ibias RRIO buffer -> SAR + external reference.
Ratiometric-to-+3V3 dead (+/-5 % rail = +/-250 mV vs a 5 mV budget, and nothing
cancels:  the source is not excited by that rail).  Bufferless dead by
t = (N+1)*ln2*(Rs+Rss)*Cs against the >=100 kohm input-impedance requirement.
- Leads:  MCP3201 (C49274, separate VREF pin) / ADR4525A (C403698, 300 mV
dropout
leaves 335 mV at a 3.135 V rail) / LT5400B (C1739858) or a 0.01 %-2 ppm
discrete
pair / OPA333 (C30878).  MCP3202 disqualified (VDD and VREF share one pin in
SOIC-8); every >=16-bit-ACCURACY SAR in stock needs >=4.5 V AVDD.
- Rail +3V3 ‘direct‘ from J2, 11 mA, 36 mW, no thermal constraints, no
AVDD/DVDD
ferrite (one net, 100 nF per pin, unsplit pour).
- Rulings recorded:  RSS claim (4.8 mV) with the 10.6 mV worst-case published
beside
it; no second rail; Extended tier for all four accuracy parts; non-ratiometric;
input range must include 0 V; ADI-branded only for AD8605-pattern parts.
- Open for P2:  source impedance and divider Rtot are ONE number pair and the
answered specs are jointly impossible at 0.1 % (1 kohm into 100 kohm = 1 %).
Matched networks cap near 100 kohm/element - network-grade matching and
megohm-grade input impedance cannot both be had.
```

**P2**

```
# P2 Architecture + coverage research - digest

- Chain:  screw terminal -> 5x 200k 0.02%/10ppm equal string tapped 3:2 (K =
0.400,
Rtot 1.00 Mohm, guarded tap) -> RRIO CMOS buffer -> ADS8326-class 16-bit SPI
SAR,
VREF 2.048 V from an ADR4520-class reference.  FS 5.12 V. 2L JLC2313_1.6,
~$30/board.
- Budget RSS 3.23 mV @25 C / 3.36 mV over 0-50 C vs 5.0/12.0; worst case
8.35/10.29.
Claimed on RSS, worst case published beside it.
- Converter overruled from P1:  MCP3201 gain error +/-5 LSB MAX = 6.10 mV =
1.22x the
whole budget.  LSB-denominated DC specs shrink 16x from 12 to 16 bits.
- Q9 narrowed (owner-visible):  source <= 200 ohm, board 1.00 Mohm, both on
silk.  As
answered the two specs were jointly impossible at 0.1 % (Rs/Rtot = 1 % = 50
mV).
- Key move:  -IN is a dedicated remote sense to the string’s bottom node.  A
divider
passes ground offset at UNITY while dividing signal by K, so 1 mV = 2.5 mV
input-referred; the sense line cancels it.  sheets.md said "-IN = GND" and was
fixed.
- Buffer OPA320-class:  OPA333 fails at t_acq 9 us (GBW floor 849 kHz vs 350
kHz; own
settling 40.4 us).  A buffer AT ALL is unconditional - switched-cap R_eq =
1/(f_s*C_SH) ~ 2.08 Mohm vs 240 kohm Thevenin is ~10 % gain error.
- Series R at the converter rail entry REINSTATED: the datasheet’s own
recommended
circuit shows it, so it is required support, not excluded filtering.
```

```
- Research:  4 tasks, 16 sources, 32 verified records + 4 checklists, 3
correction
rounds each.  Coverage now provisional everywhere except B1
constraints-emission,
recorded as a designed-under gap.  Promotion deferred (shared library, live
siblings).
```

**P3**

```
# P3 Parts + Library - digest

- BOM 16 lines / 20 refdes, 7 Basic + 9 Extended, $155.16 at qty 5
($31.03/board);
U1 + U2 are 84 % of it.  U1 single-source with an explicitly EMPTY alternates
list.
- All three verification items CLOSED from primary sources:  acquisition is a
CLOCK-CYCLE COUNT (tSMPL 4.5-5.0 DCLOCKs) -> OPA320 final, OPA333 demoted; the
-IN
window is an ABSOLUTE -0.3/+0.5 V constant; ADR4520 still Rev 0 confirmed via a
second independent channel, so the 87 mV margin stands.
- 3 extractions schema-valid; library 13 parts / 11 footprints loading clean;
-IN
typed as a signal pin, not power.
- BOARD-KILLER CAUGHT: J1 drilled 1.30 mm where the vendor specifies 1.50 mm.
These
terminals carry ~1.0 mm SQUARE pins (1.41 mm diagonal) - the part could not
have
entered its holes, and 1.30 mm CLEARS the annulus floor so every check passed.
Corrected and re-verified.
- MAJOR OPEN RISK for P8:  TI's own circuit for this converter+amplifier pair
warns
single-supply "loses codes near ground" and our span starts at 0 V. DC says ~61
uV;
the transient is unproven.  `zero-scale-swing` + `acquisition-settling` decide
it.
A -0.3 V generator is pre-ruled datasheet-required support but still a second
rail.
- OPA320 passes the isolation screen OPA333 failed and publishes the flat 90
ohm R_O
the method assumes; its settling is uncharacterised at our supply and load.
- Coverage 3/3 part slots COVERED; B2/B3/B4 provisional (maturity, not
mapping);
B1 gap on constraints-emission only.  No research spent, 2 tasks remain.
```

**P4**

```
# P4 Schematic - digest

- ONE schematic agent, FLAT root sheet (a child-sheet label exports as
/<sheet>/<LABEL> and silently unhooks every constraints.json entry - the
defect that
cost lumina-par).  24 components, 14 canonical nets, 61/61 pins connected.
- Gate `erc` PASS 0/0 (2 attempts; commits 1befa1e then d1738e6).
netlist_audit exit 0
with one accepted warning (`power_no_consumers` on VREF - the REF pin is typed
`passive` in the pulled symbol; not hand-retyped because lib_pin_types.py would
silently revert U2's five no_connect pins).
- Adversarial review 1 error / 5 warnings.  E1 was the important one and it
overturned
```

```
my own guidance:  -IN on GND with the Kelvin intent carried by PLACEMENT does
not
survive the pour, because the pour TERMINATES pin 3 locally and the sense node
stops
being the string bottom at all.  Fixed by giving it its own net /AGND_SENSE =
{U1.3, R5.2, R8.1} with a single 0 ohm tie R8 - netlist-enforced, so every gate
now
checks it.  Residual is the J1-return-to-string-bottom offset at sub-1 uV.
- Warnings actioned:  the error budget had been computed on the OPA333 and was
restated
for the OPA320 actually fitted; the host power-up wait rose 10 -> 50 ms
(charging
49.2 uF at the reference's 10 mA limit is ALREADY 7-10 ms); J2's destructive
mis-mate stated; C7's floor to be sized on GUARANTEED minimum at P8.
- The reviewer parsed the s-expression and the netlist directly rather than
trusting
the rendered PDF - which is how E1 was found at all.
```

**P5**

```
# P5 Board Setup - digest

- ‘board_init‘ PASS: parity 0, setup_violations 0, 41 unconnected (expected,
unrouted).
25 components, 20 nets.  Stackup ‘JLC2313_1.6‘, 2 layers, 1 oz, profile
‘2layer_1oz‘.
- Outline ‘auto‘ -> provisional 63.95 x 66.03 mm.  Deliberately generous:
geometry is an
OUTPUT at this binding, and P6 places to the canonical layout, not to fill
this.
- First run FAILED on duplicate footprints:  the schematic already carries
H1-H4, and
‘--mounting-holes 4‘ added its own.  Re-run with ‘--mounting-holes 0‘.
- Fab floors written at ERROR severity:  clearance 0.127, track 0.127, via dia
0.6,
hole 0.3, annulus 0.15, hole-to-hole 0.5, edge clearance 0.3.
- ‘rules_gen‘ PASS, capability_class ‘2layer_1oz‘, 12 rules.  NO netclass split:
every
rail's IPC-2152 width (max 5.5 um for +3V3 at 11 mA) is far under the 0.2 mm
Default,
so all four power nets stay Default.  No HV rules - 5 V is the highest net on
the board.
- Sidecars ‘constraints.json‘ + ‘decoupling.json‘ now sit beside the board, as
every
later gate resolves them from the board's own directory.
```

**P6**

```
# P6 Placement - digest

- Hand-placed canonical layout; EVERY anneal candidate rejected.  Root cause
recorded:
place_anneal RE-DERIVES satellite slots from place_seed rather than inheriting
the
placement handed to it, so a 0.7 % hpwl win cost a 60 % violation of a declared
decoupling distance (C3 3.97 mm against a 2.5 mm limit).  hpwl 327.23 -> 223.47
mm.
- Gate ‘place‘ PASS 0/0; route probe completion 1.00 (41/41); 25/25 front,
25/25 locked
```

```
(the guard room, tap stub, sense tie and string order are specifications no
gate can
re-verify - locking is what stops a later pass optimising them away).
- Geometry EARNED: 63.95 x 66.03 provisional -> 54.750 x 34.920 mm, 1911.9 mm2.
The
J1-to-R1 span decomposes as 4.5 mm connector body + 2.0 mm screwdriver access +
0.8 mm pad approach = 7.3 mm.  The earlier 16.5 mm was not merely unearned, it
was
WORSE: every record touching that net says leakage and pickup scale with
length.
H1/H2 - not J1 - had been defining the west edge.
- Silk 0 warnings after silk_place (2 refdes moved, no residual).
- Handed to P7:  the SPI fan needs exactly 2 crossings (J2 CS,SCLK,DOUT vs
converter
CS,DOUT,SCLK is a 3-cycle), and the guard ring cannot close through the 0.35 mm
U3 pin2/pin3 channel, so it must encircle V- together with the input pin.
```

**P7**

```
# P7 Routing - digest

- 2-layer chain, one deliberate substitution:  'route_critical' SKIPPED (no
diff pairs,
no RF, no high-current net, and its 0.3 mm POWER_WIDTH_FLOOR would have forced
widths)
and replaced by a hand pre-route of the analog-critical nets.  planes_gen ->
route_auto (rung 1, 4 passes, completion 1.00, 73 SES items) -> stitch_vias DRY
ONLY
-> plane_repair (0 splits) -> route_cleanup SKIPPED (2L pour board, S14).
- Gate 'drc_routed' PASS 0/0.  GND pour 1745.23 mm2, single polygon.
- Guard ring CLOSED and PROVEN geometrically, not eyeballed:  the whole
/AIN_BUF F.Cu
net is one polygon with one interior hole that contains() all /AIN_DIV copper,
min
ring width 0.612 mm, guard-to-node gap 0.127 mm.  It closes ONLY at the 0.127
mm
floor - at 0.2 mm the south leg is culled, the pour drops 19.03 -> 8.68 mm2 and
there
are ZERO holes, yet it still fills and still looks like a ring in a render.
- +3V3 widened 0.2 -> 0.5 mm per constraints (impedance to the decoupling caps,
not
current - my brief had argued from current and was wrong), with a 'Power3V3'
netclass
added so a future re-route cannot silently revert it.  R 0.108 -> 0.043 ohm.
- Two pipeline defects found:  stitch_vias models hole-to-hole CENTRE spacing
while
KiCad measures EDGE to edge (unsafe band 0.5-0.8 mm for 0.3 mm drills); and
geom._refill_copy stages the .kicad_pro but NOT the .kicad_dru, so --verify-fill
refills without the 0.127 mm floor and reports a false stale fill on EVERY
board.
```

**P8**

```
# P8 Verification - digest

- Gates:  'sim' PASS (6 benches / 47 bounds), 'drc_routed' PASS 0/0, 'verify'
PASS with
ONE waiver.  kicad-cli DRC 0 violations, 0 unconnected, 0 parity.
- SIM closed the board's largest open risk BY MEASUREMENT: zero-scale swing is
4.50 uV
```

```
= 0.090 ADC codes against the vendor's "loses a small number of codes" warning,
and
the sampled value recovers to within 10 nV of ground in 1.24 us of a 9.0 us
window.
The pre-authorised -0.3 V generator is NOT needed.  C7 stays 1 nF (guaranteed
950 pF
settles 115x inside budget).  Seeding the REJECTED OPA333 as a defect moved
settling
0.136 -> 4509 uV, independently re-auditing that rejection by a different
method.
- Return path:  /SCLK and /DOUT cleared by rebuilding the /CS crossing as a
PERPENDICULAR
VIA PAIR so both crossings fall inside the vias' 0.95 mm excision disks.  /CS
residual
0.20 mm2 / 0.03 mm WAIVED (2-layer:  a B.Cu trace has no adjacent GND plane, so
its
coplanar return is outside the checker's model; and J2 vs converter pin order
is a
3-cycle, so the crossing is topologically forced).  Refused to shorten the
tunnel
0.09 mm to close it - no physics, ~0.04 mm of DRC margin.
- Board review (fresh context) 5 errors / 5 warnings.  ALL FIVE ERRORS FIXED:
E1 J1 was BACKWARDS - throats faced into the board; rotated, re-routed (a 180
deg
rotation swaps the pads).  E5 GND inside the guard ring - fixed by re-routing
the
GUARDED net's spanning tree, not the intruder:  1.20 -> 0.52 mV. E2/E3/E4 silk:
J2 pin-1 mark outside the body, J1 SIG/GND, and both impedance numbers.
- Guard ring re-proved CLOSED after every refill (1 polygon, 1 hole containing
all
/AIN_DIV, min width 0.612 mm) - and it closes ONLY at the 0.127 mm floor.
- Board 54.750 x 34.920 mm, 1911.9 mm2.  A 17 um re-fit was refused as
sub-noise.
```

**P9**

```
# P9 DFM - digest

- `fab_export` PASS: 9 layers, Protel-extension gerbers + drill, zipped
(`fab/bb-adc_gerbers.zip`, sha recorded), plus the position export.
- `bom_cpl` PASS: BOM of record + JLC upload pair + CPL. 21 parts, ALL placed,
`bom_complete` true, class split 21 `smt_placed` / 4 `board_feature` (the
mounting
holes).  No missing LCSC, no unsourced, no off-LCSC, no unplaced SMT, no qty
mismatch.
**2 rotation corrections applied** from `jlc_rotations.csv` - the catcher for a
polarized part mounted backwards, which net-level parity is blind to by
construction.
- Gate `dfm` PASS 0/0.  Note what it graded:  gerbers RE-EXPORTED to scratch
from the
board, plus the sibling schematic as the CPL polarity oracle and parts.json for
the
BOM leg.  The shipped zip is an ordering artifact and is NOT a gate input.
- Board 54.750 x 34.920 mm, 2 layers, 1 oz, HASL, no impedance control - the
cheapest
class JLC sells.  ~$31/board of parts at qty 5, 84 % of it U1 + U2.
- Stops here by mode:  `block-only` defaults end at P9 with the fab package.
Ordering is
a separate owner decision and no money has been committed.
```

## Gate attempt history

| Timestamp | Gate | Status | Failing/Total |
|---|---|---|---|
| 2026-08-17T21:30:22 | erc | pass | 0/0 |
| 2026-08-18T08:52:06 | erc | pass | 0/0 |
| 2026-08-19T02:48:09 | place | pass | 0/0 |
| 2026-08-19T02:48:14 | place | pass | 0/0 |
| 2026-08-19T02:50:07 | sim | pass | 0/1 |
| 2026-08-19T02:50:42 | sim | pass | 0/1 |
| 2026-08-19T02:50:55 | place | pass | 0/0 |
| 2026-08-19T02:51:31 | sim | pass | 0/1 |
| 2026-08-19T02:52:42 | place | pass | 0/0 |
| 2026-08-19T02:53:14 | sim | pass | 0/1 |
| 2026-08-19T02:54:55 | place | pass | 0/0 |
| 2026-08-19T03:00:09 | place | pass | 0/0 |
| 2026-08-19T03:25:28 | drc_routed | pass | 0/0 |
| 2026-08-19T03:28:03 | drc_routed | pass | 0/0 |
| 2026-08-19T03:33:23 | drc_routed | pass | 0/0 |
| 2026-08-19T03:34:44 | drc_routed | pass | 0/0 |
| 2026-08-19T03:35:21 | verify | fail | 2/4 |
| 2026-08-19T03:46:23 | drc_routed | pass | 0/0 |
| 2026-08-19T03:46:24 | verify | fail | 1/2 |
| 2026-08-19T03:46:31 | verify | fail | 1/2 |
| 2026-08-19T03:49:10 | drc_routed | pass | 0/0 |
| 2026-08-19T03:49:10 | verify | fail | 1/2 |
| 2026-08-19T03:50:27 | verify | pass | 0/2 |
| 2026-08-19T04:42:38 | drc_routed | pass | 0/0 |
| 2026-08-19T04:42:46 | verify | pass | 0/2 |
| 2026-08-19T04:44:50 | drc_routed | pass | 0/0 |
| 2026-08-19T04:44:51 | verify | pass | 0/1 |
| 2026-08-19T04:52:53 | drc_routed | pass | 0/0 |
| 2026-08-19T04:52:54 | verify | pass | 0/1 |
| 2026-08-19T04:53:50 | drc_routed | pass | 0/0 |
| 2026-08-19T04:53:50 | verify | pass | 0/1 |
| 2026-08-19T04:56:09 | dfm | pass | 0/0 |
| 2026-08-19T04:56:31 | dfm | pass | 0/0 |
| 2026-08-19T04:57:36 | erc | pass | 0/0 |
| 2026-08-19T04:57:37 | place | pass | 0/0 |
| 2026-08-20T00:46:04 | erc | pass | 0/0 |
| 2026-08-20T00:46:26 | drc_routed | pass | 0/0 |

## Issues

| Id | Phase | Kinds | Severity | Status |
|---|---|---|---|---|
| 1 | P8 | corridor_void | error | fixed |

## Human checkpoints

| Id | Phase | Status | When | Note |
|---|---|---|---|---|
| 1 | P2 | approved | 2026-08-16T21:31:27 | H1 presented as report-and-proceed per the owner's mid-run delegation ('make decisions yourself based on what will give the best learning for the basic system'). Nothing here is blocking except the Q1 safety envelope re-confirmation and the Q9 source-impedance narrowing, both surfaced explicitly. Design doc: reports/design_doc/bb-adc-design-doc.pdf |
| 2 | P4 | approved | 2026-08-18T08:52:52 | H2 report-and-proceed per owner delegation. ERC gate PASS (2 attempts, commit d1738e6). Review: 1 error FIXED (E1 Kelvin sense now netlist-enforced via /AGND_SENSE + R8 0R tie), 5 warnings dispositioned in reports/erc-waivers.md. Architecture docs patched to match. |
| 3 | P6 | not reached | | |
| 4 | P8 | approved | 2026-08-19T04:54:08 | H4 report-and-proceed per owner delegation. verify PASS (1 waiver: /CS return path, out-of-model on 2L + topologically forced). Review 5 errors: E1 J1 backwards FIXED (rotated, re-routed), E5 guard leakage FIXED (guarded-net reroute, 1.20 -> 0.52 mV), E2/E3/E4 silk FIXED. W1 root cause fixed. Board 54.75x34.92 mm earned. Commits d1738e6 erc, 4331069 sim, 369db07 drc_routed, 5c6c692+e877d11 verify. |
| 5 | P10 | not reached | | |

### Generation cost

*Not recorded (no reports/cost.json).*

## 10 Artifact Index

| File | Size | Note |
|---|---|---|
| state.json | 194,208 B | |
| kicad/PCB-0014-A_bb-adc.kicad_pcb | 150,002 B | |
| kicad/PCB-0014-A_bb-adc.kicad_sch | 106,881 B | |
| fab/PCB-0014-A_bb-adc_gerbers.zip | 23,905 B | |
| fab/BOM.csv | 600 B | |
| fab/CPL.csv | 712 B | |
| brief/brief.md | 742 B | |
| requirements.md | 23,802 B | |
| architecture/blocks.md | 38,927 B | |
| architecture/constraints.json | 28,600 B | |
| architecture/power_tree.md | 12,951 B | |
| architecture/sheets.md | 15,417 B | |
| architecture/stackup.md | 5,012 B | |
| reports/board_edit-fit.json | 1,922 B | |
| reports/board_init.json | 1,519 B | |
| reports/bom_cpl.json | 12,234 B | |
| reports/drc-postfit.json | 31,427 B | |
| reports/drc-refill.json | 563 B | |

| File | Size | Note |
|---|---|---|
| reports/erc-waivers.md | 2,572 B | |
| reports/erc.json | 448 B | |
| reports/fab_export.json | 2,134 B | |
| reports/fix-dispatch-verify.json | 616 B | |
| reports/gate-dfm.json | 2,022 B | |
| reports/gate-drc_routed.json | 1,668 B | |
| reports/gate-erc.json | 837 B | |
| reports/gate-place.json | 1,421 B | |
| reports/gate-sim.json | 1,319 B | |
| reports/gate-verify.json | 4,958 B | |
| reports/netlist_audit.json | 1,399 B | |
| reports/place-final.ops.json | 4,196 B | |
| reports/place.json | 5,279 B | |
| reports/review-board.json | 10,788 B | |
| reports/review-board.md | 20,130 B | |
| reports/review-schematic.json | 5,180 B | |
| reports/review-schematic.md | 20,553 B | |
| reports/route-probe.json | 287 B | |
| reports/rules_gen.json | 1,362 B | |
| reports/schematic.pdf | 102,209 B | |
| reports/silk_place.json | 3,104 B | |
| reports/sim.json | 7,085 B | |
| reports/top.net | 48,090 B | |
| reports/verify-waivers.json | 1,775 B | |